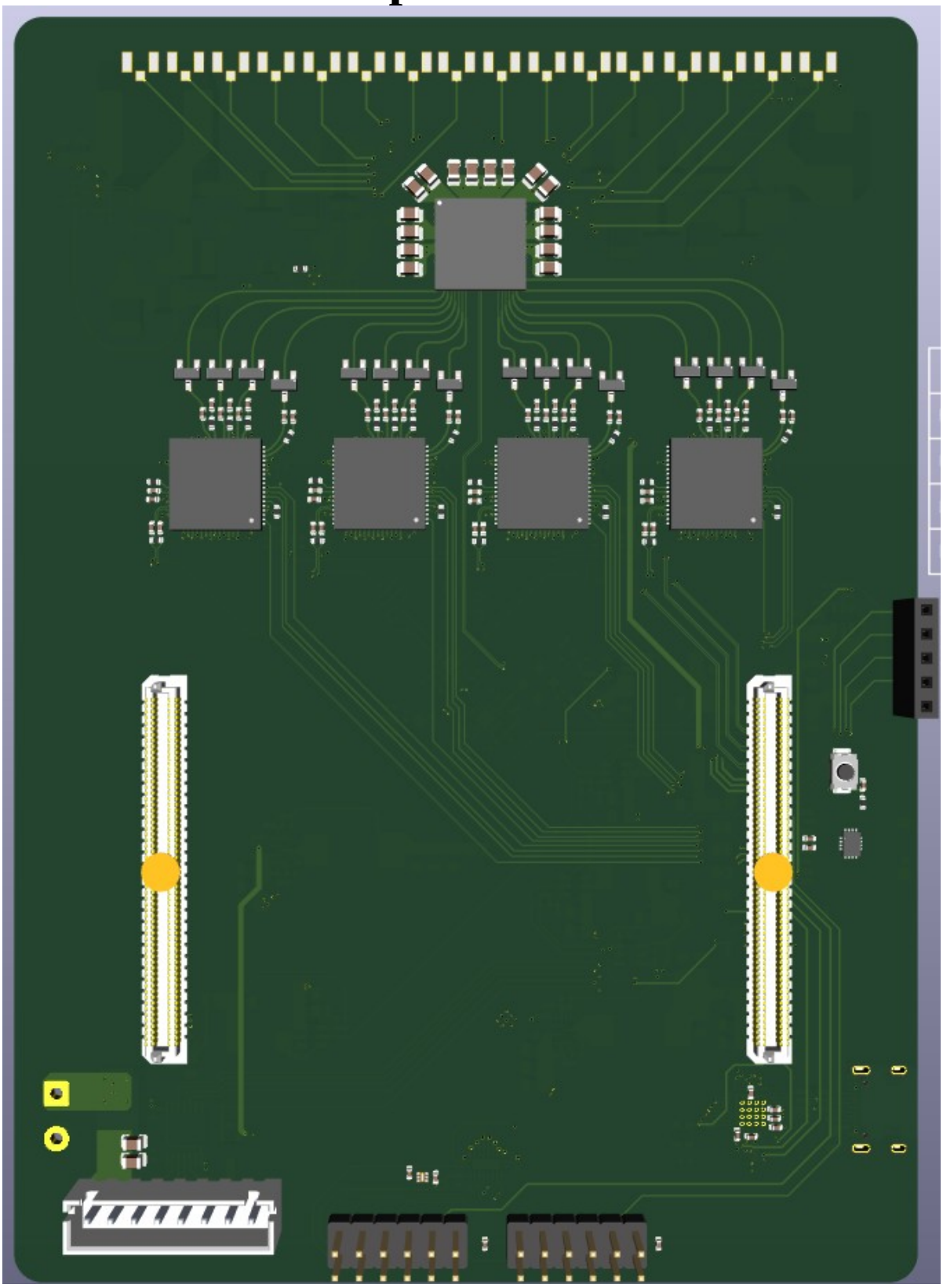
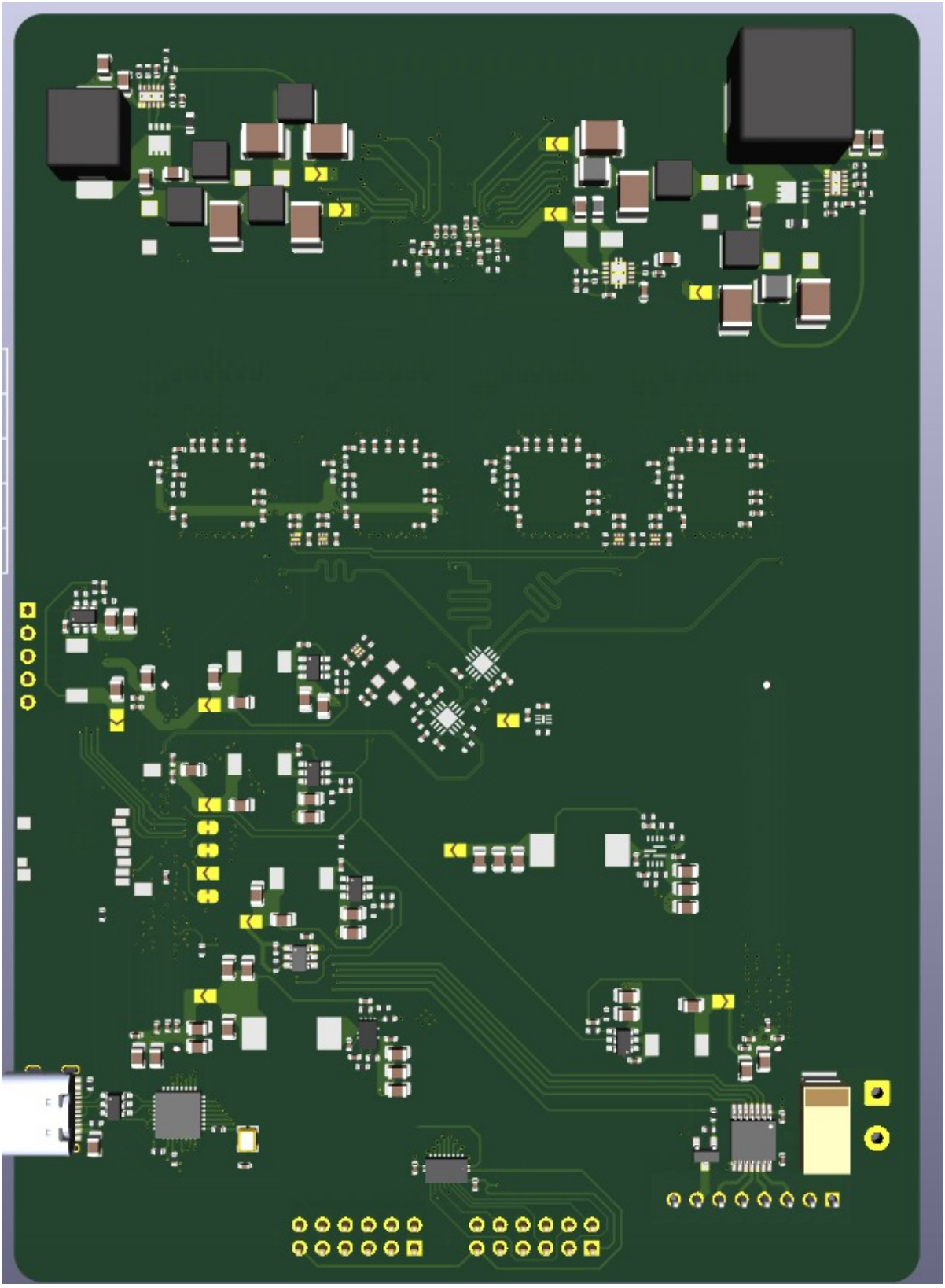
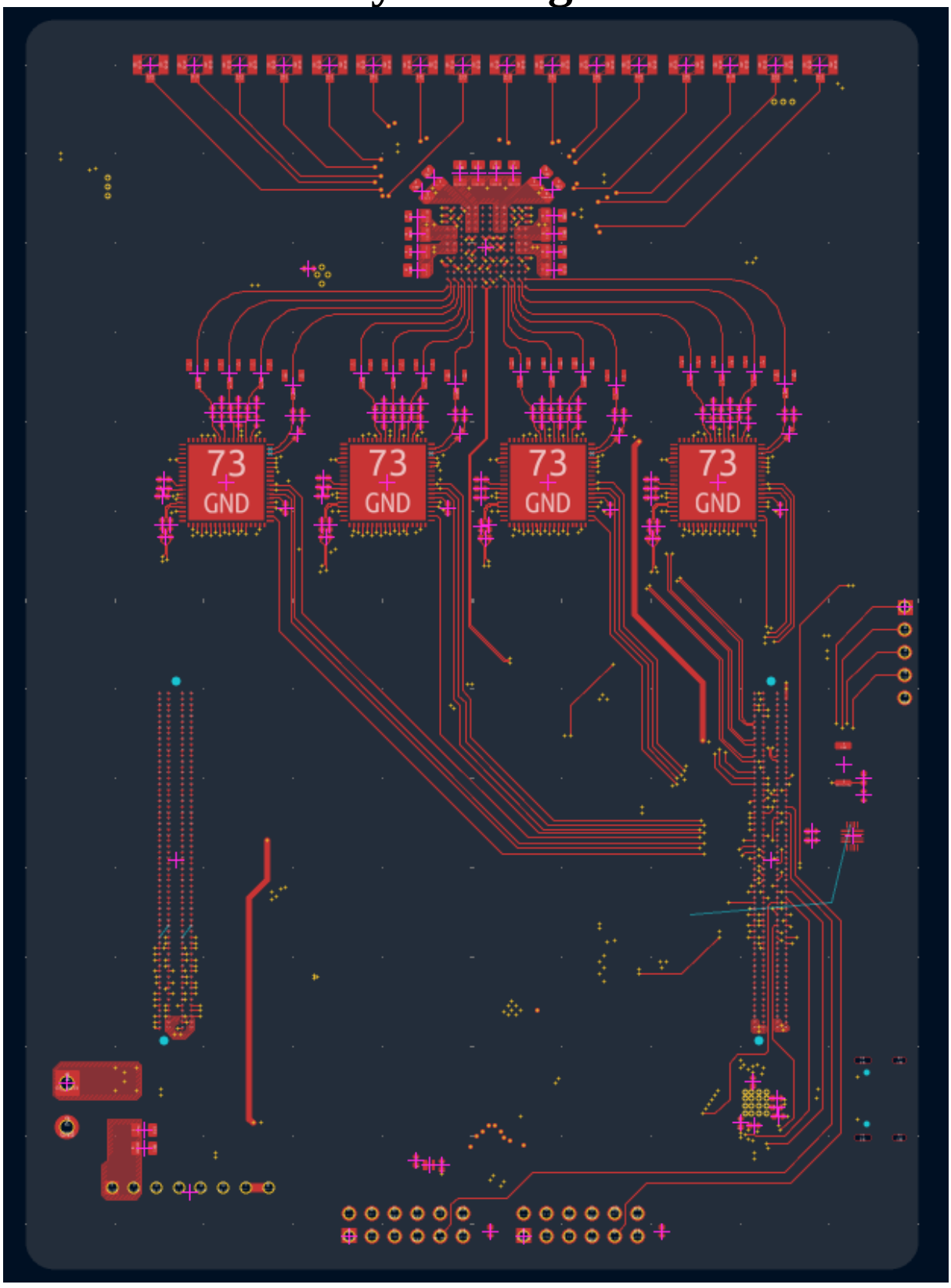
Top 3D View



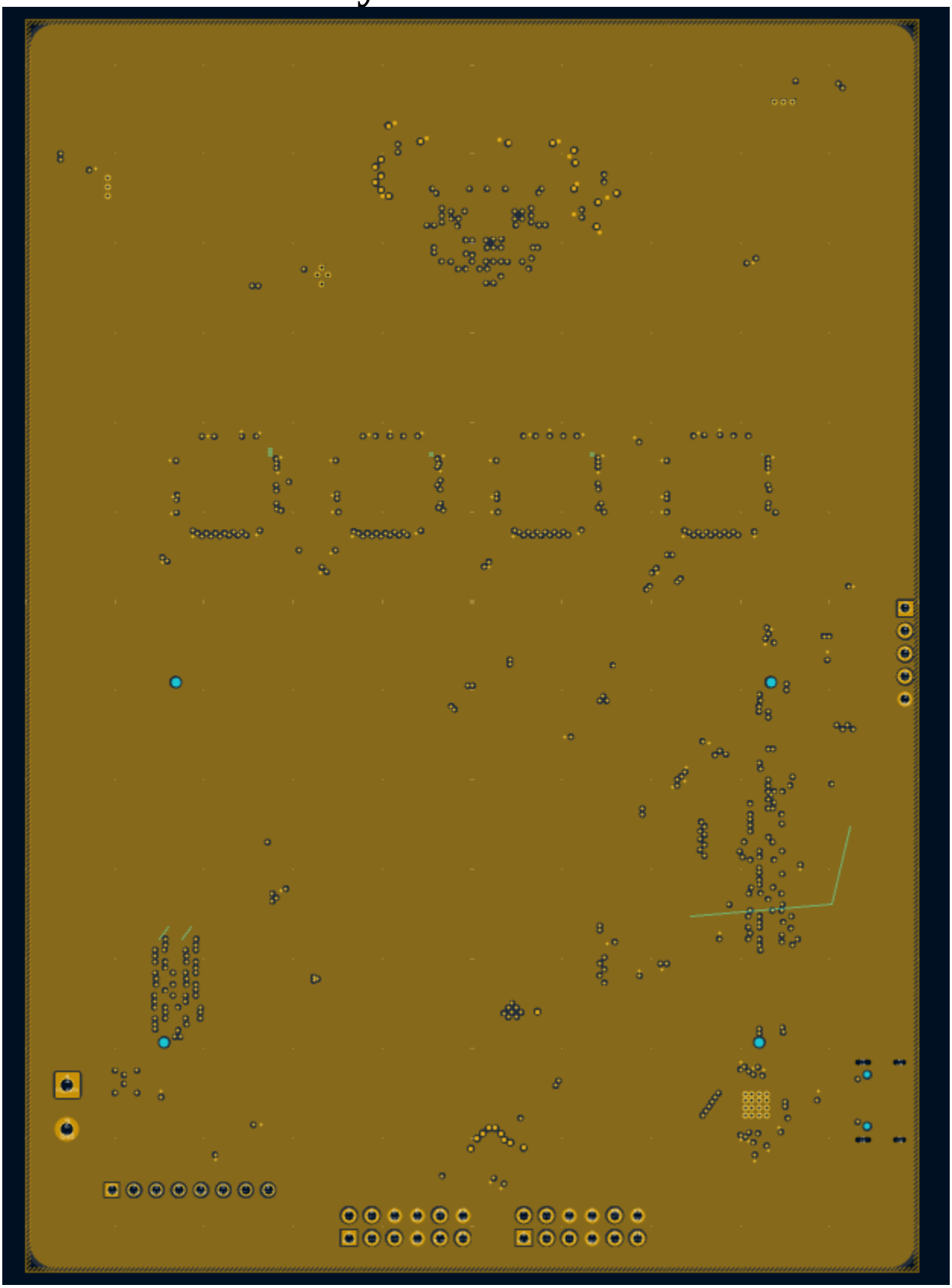
Bottom 3D View



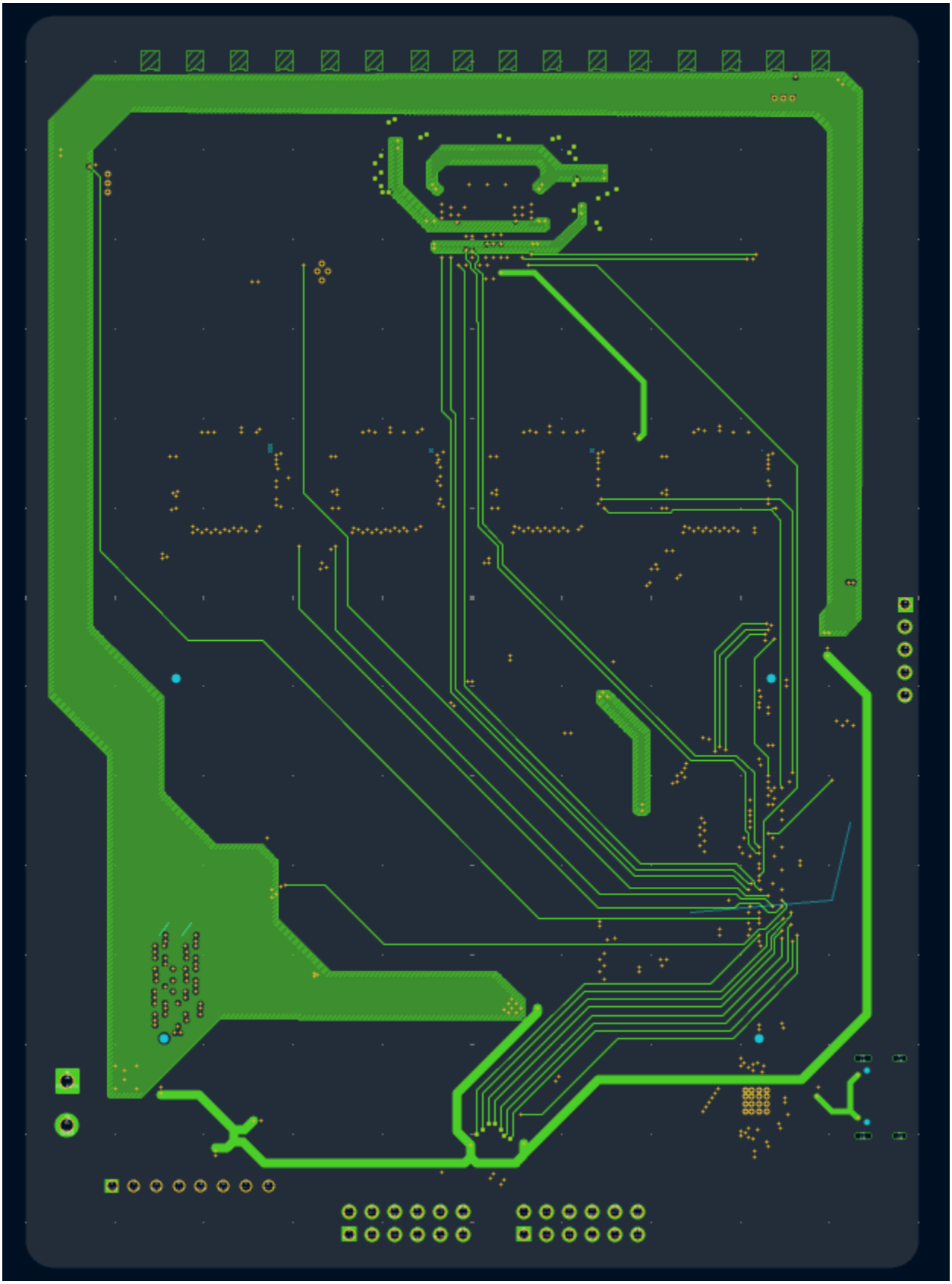
Layer 1: Signal



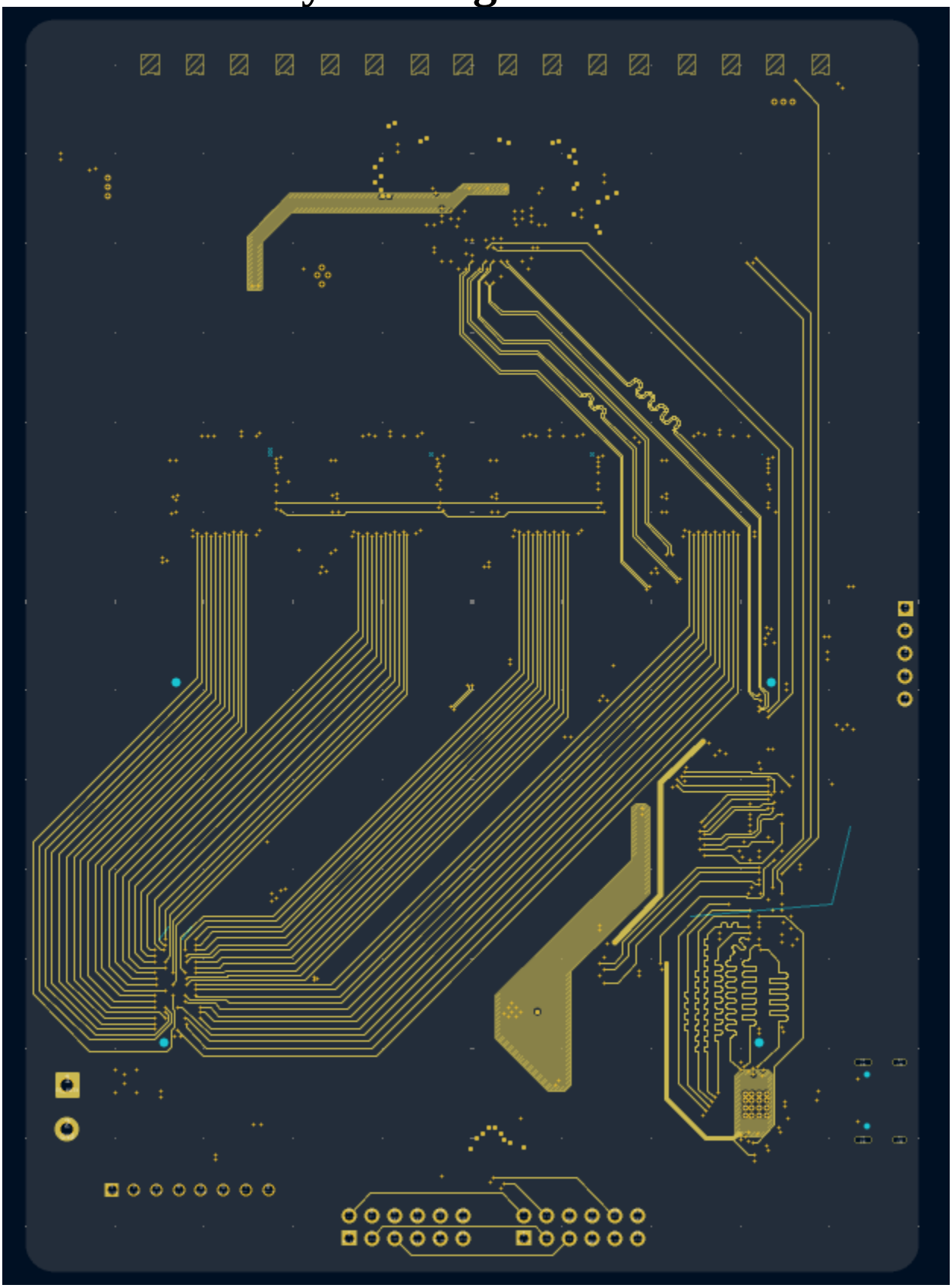
Layer 2: Ground



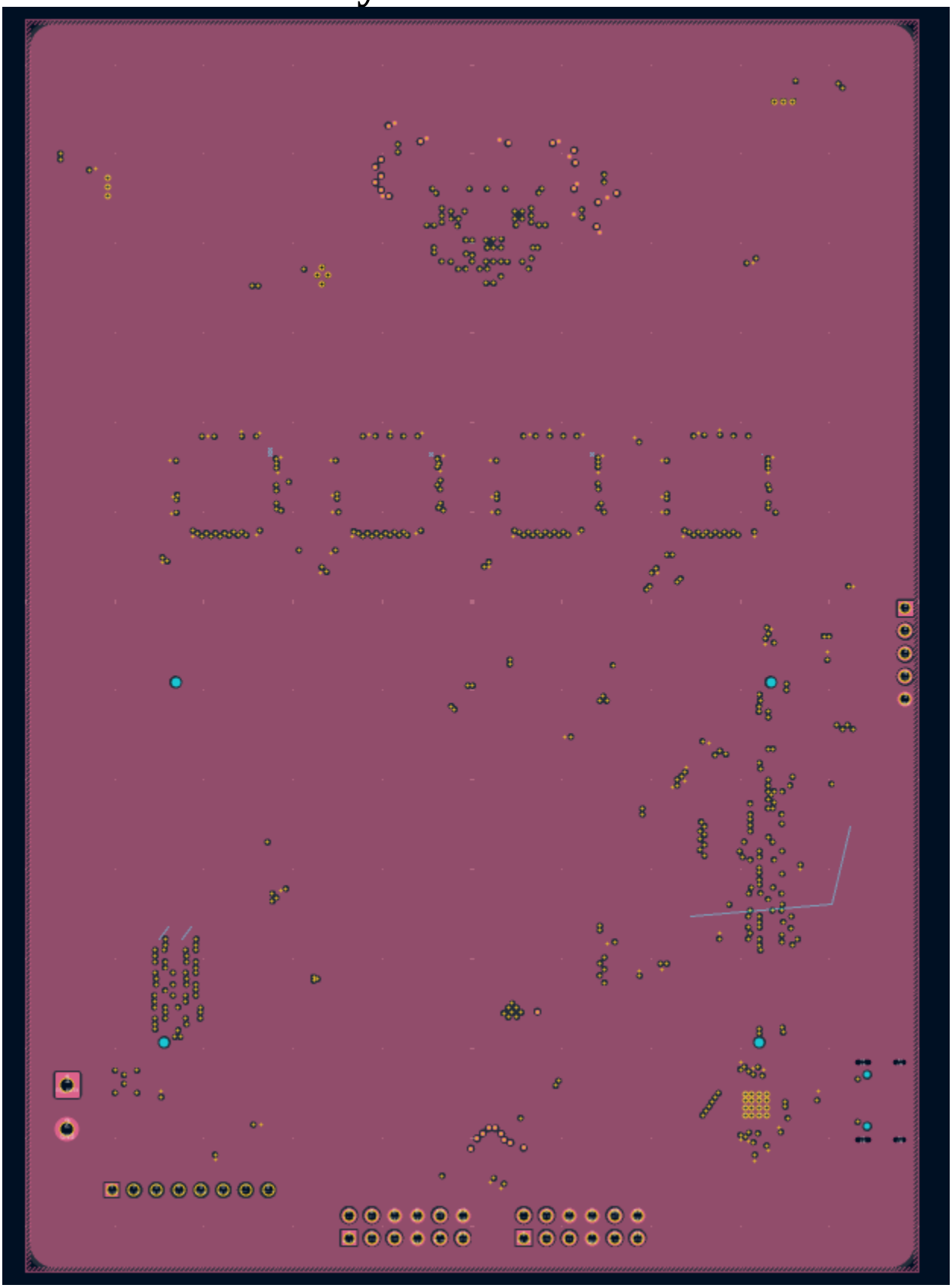
Layer 3: Signal/Power



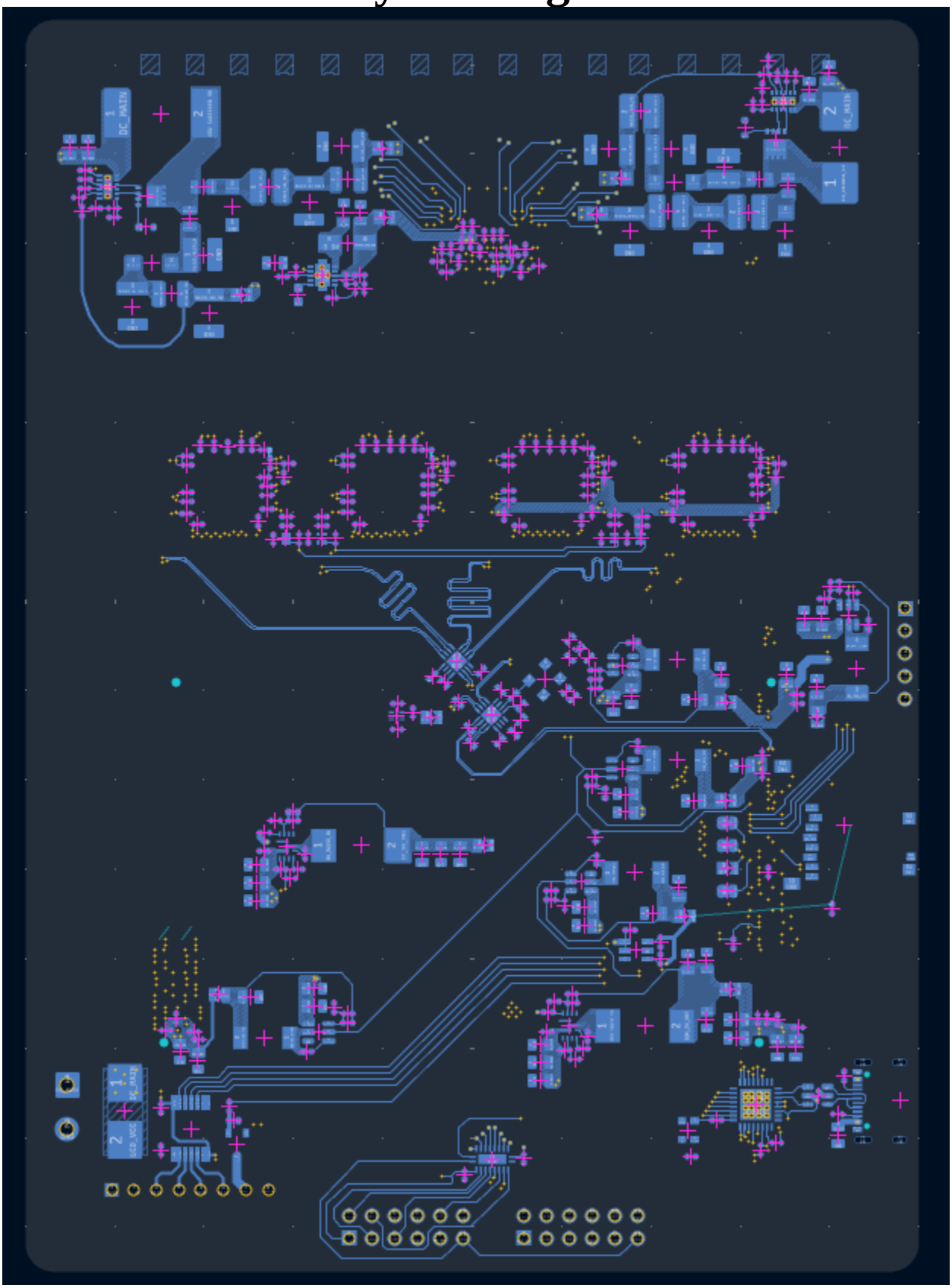
Layer 4: Signal/Power

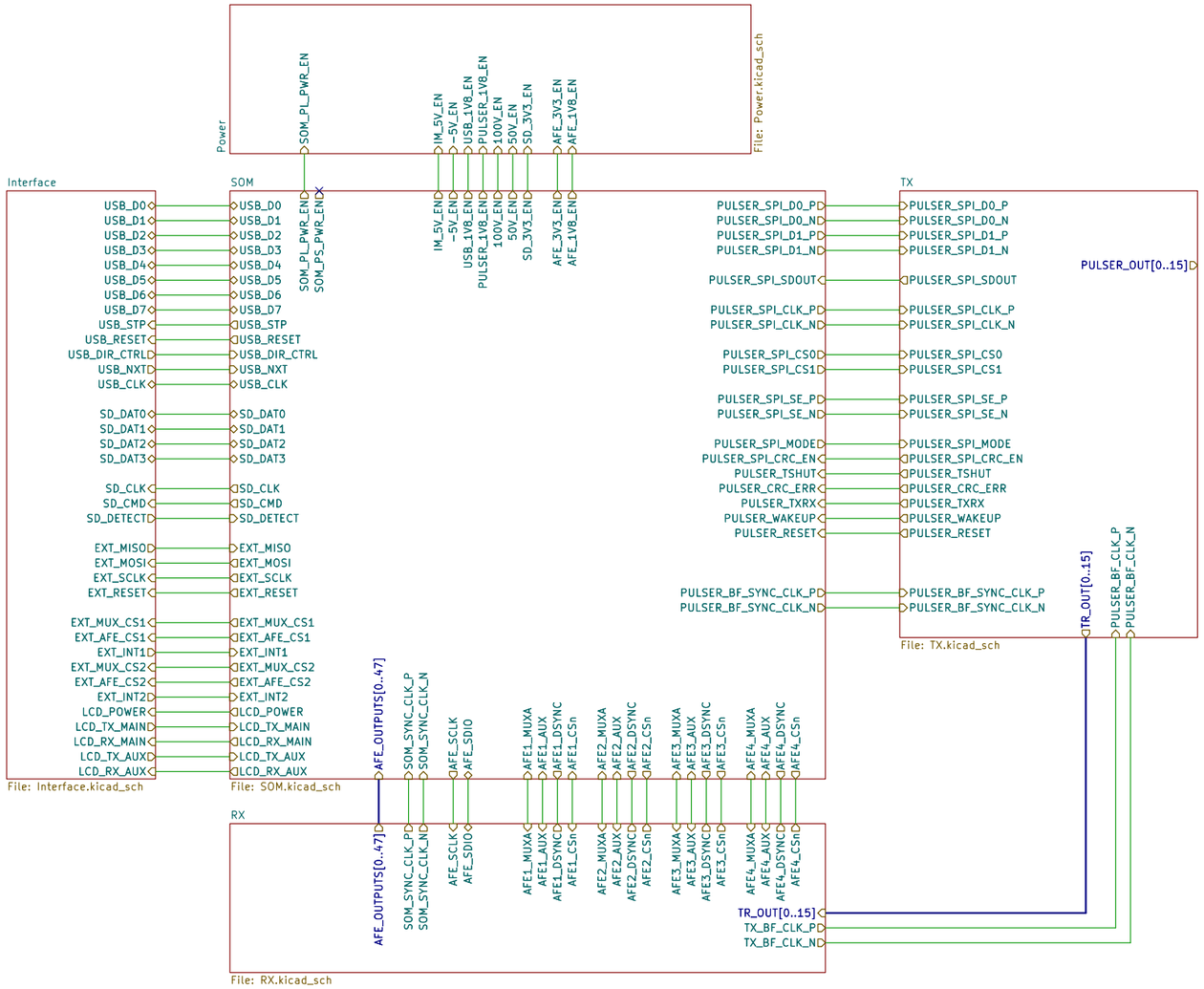


Layer 5: Ground



Layer 6: Signal





Power Adapter	12-24V (5A)	SOM Buck Converter	5V 4A (1.6A @ 12V)	Main SOM Supply (4A)	
		Intermediate Voltage Buck Converter	5V 2A (0.8A @ 12V)	Pulser Main Voltage (5V, 0.016A)	
				Pulser Negative Voltage (-5V, 0.023A)	
				AFE 1 & 2 Main Supply (3.3V, 0.26A)	
				AFE 1 & 2 Logic Supply (1.8V, 0.3A)	
				AFE 3 & 4 Main Supply (3.3V, 0.26A)	
				AFE 3 & 4 Logic Supply (1.8V, 0.3A)	
				Clock LDO Regulator 1	3.3V 0.1A
				Clock LDO Regulator 2	1.8V 0.1A
				External Interfaces LDO Regulator	3.3V ?A Up to 2.5A
		HDA Buck Converter	3.3V 1A (0.3A @ 12V)		
		HDB & HDC Buck Converter	3.3V 1A (0.3A @ 12V)		
		HPA Buck Converter	1.8V 1A (0.3A @ 12V)		
		USB & Pulser Buck Converter	1.8V 0.064A		
		SD Card Buck Converter	3.3V 1.2A (0.35A @ 12V)		
		+/-100V HV Converter	+/-100V 0.0005A (0.004A @ 12V)		
		+/-50V HV Converter	+/-50V 0.3A (1.2A @ 12V)		

TODO:
- **DETAIL THE INDUCTORS' FOOTPRINTS AT THE END**
- **Add the input power decoupling capacitors & necessary input power protetion**

U201A

PULSER_SPI_CLK_PD — H6 SCLKP
PULSER_SPI_CLK_ND — J6 SCLKM
PULSER_1V8
PULSER_SPI_SDOUT — H12 SDOUT
R202 10K
PULSER_SPI_DO_PD — H9 SDATAP_0
PULSER_SPI_DO_ND — J9 SDATAM_0
Only place resistors R1 & R2 if data line 1 is not used.
PULSER_SPI_D1_PD — H7 SDATAP_1
PULSER_SPI_D1_ND — J7 SDATAM_1
R205 10K
PULSER_SPI_SE_PD — H4 SENP
PULSER_SPI_SE_ND — J4 SENM
PULSER_SPI_MODE — G5 SPI_MODE
GND
PULSER_BF_SYNC_CLK_PD — L7 TR_BF_SYNCP
PULSER_BF_SYNC_CLK_ND — L6 TR_BF_SYNCM
PULSER_BF_CLK_PD — M7 BF_CLKP
PULSER_BF_CLK_ND — M6 BF_CLKM
PULSER_SPI_CS0 — F7 CHIP_SEL0
PULSER_SPI_CS1 — G7 CHIP_SEL1
PULSER_TXRX — H11 TX_RXZ
PULSER_SPI_CRC_EN — G4 EN_CRC
PULSER_TSHUT — H1 TSHUT
PULSER_WAKEUP — J12 WAKE_UP
PULSER_RESET — J1 RESET
PULSER_CRC_ERR — J2 CRC_ERR
R201 10K
R203 10K
GND
GND

OUT_1 A12 PULSER_OUT0
OUT_2 B12 PULSER_OUT1
OUT_3 C12 PULSER_OUT2
OUT_4 A11 PULSER_OUT3
OUT_5 B11 PULSER_OUT4
OUT_6 C11 PULSER_OUT5
OUT_7 A10 PULSER_OUT6
OUT_8 B10 PULSER_OUT7
OUT_9 A1 PULSER_OUT8
OUT_10 B1 PULSER_OUT9
OUT_11 C1 PULSER_OUT10
OUT_12 A2 PULSER_OUT11
OUT_13 B2 PULSER_OUT12
OUT_14 C2 PULSER_OUT13
OUT_15 A3 PULSER_OUT14
OUT_16 B3 PULSER_OUT15

PULSER_OUT[0..15]

HV Pulser Outputs

RX_1 L12 TR_OUT0
RX_2 M12 TR_OUT1
RX_3 L11 TR_OUT2
RX_4 M11 TR_OUT3
RX_5 L10 TR_OUT4
RX_6 M10 TR_OUT5
RX_7 L9 TR_OUT6
RX_8 M9 TR_OUT7
RX_9 L4 TR_OUT8
RX_10 M4 TR_OUT9
RX_11 L3 TR_OUT10
RX_12 M3 TR_OUT11
RX_13 L2 TR_OUT12
RX_14 M2 TR_OUT13
RX_15 L1 TR_OUT14
RX_16 M1 TR_OUT15

TR_OUT[0..15]

T/R Switch Outputs

NC F5
NC F6
NC G6
NC K1
NC K12

Output:
Indicates whether the pulser is in transmit or receiver mode.

U201B

PULSER_5V — Logic 5V
E9 AVDDP_5
E8 AVDDP_5
E7 AVDDP_5
E6 AVDDP_5

PULSER_5VN — −5V
E4 AVDDM_5V
E5 AVDDM_5V

PULSER_1V8 — 1.8V
F8 AVDDP_1P8
F9 AVDDP_1P8
G8 AVDDP_1P8
G9 AVDDP_1P8

PULSER_100VN — −100V HV Supply
F1 AVDDM_HV_A
F2 AVDDM_HV_A
F12 AVDDM_HV_A
F11 AVDDM_HV_A
G1 AVDDM_HV_A
G2 AVDDM_HV_A
G11 AVDDM_HV_A
G12 AVDDM_HV_A

PULSER_50VN — −50V HV Supply
D1 AVDDM_HV_B
D2 AVDDM_HV_B
D11 AVDDM_HV_B
D12 AVDDM_HV_B
E1 AVDDM_HV_B
E2 AVDDM_HV_B
E11 AVDDM_HV_B
E12 AVDDM_HV_B

PULSER_100V — +100V HV Supply
A4 AVDDP_HV_A
A9 AVDDP_HV_A
B4 AVDDP_HV_A
B9 AVDDP_HV_A
C4 AVDDP_HV_A
C9 AVDDP_HV_A
D4 AVDDP_HV_A
D9 AVDDP_HV_A

PULSER_50V — +50V HV Supply
A5 AVDDP_HV_B
A8 AVDDP_HV_B
B5 AVDDP_HV_B
B8 AVDDP_HV_B
C5 AVDDP_HV_B
C8 AVDDP_HV_B
D5 AVDDP_HV_B
D8 AVDDP_HV_B

AVSS B6
AVSS B7
AVSS C3
AVSS C6
AVSS C7
AVSS C10
AVSS D3
AVSS D6
AVSS D7
AVSS D10
AVSS E3
AVSS E10
AVSS F3
AVSS F4
AVSS F10
AVSS G3
AVSS G10
AVSS H3
AVSS H5
AVSS H8
AVSS H10
AVSS J3
AVSS J5
AVSS J8
AVSS J10
AVSS K2
AVSS K3
AVSS K4
AVSS K5
AVSS K6
AVSS K7
AVSS K8
AVSS K9
AVSS K10
AVSS K11
AVSS L5
AVSS L8
AVSS M5
AVSS M8
GND

FLOATP_HV_A A6
FLOATM_HV_A J11
FLOATP_HV_B A7
FLOATM_HV_B H2

PULSER_TXRX
R204 10K
C201 1nF
GND

PULSER_TSHUT
R206 10K
C202 1nF
GND
Output:
Thermal shutdown flag.

PULSER_CRC_ERR
R207 10K
C203 1nF
GND
Output:
SPI error flag (CRC error).

PULSER_OUT[0..15]
PULSER_OUT0 — J201 U.FL-R-SMT-1_80_
PULSER_OUT1 — J202 U.FL-R-SMT-1_80_
PULSER_OUT2 — J203 U.FL-R-SMT-1_80_
PULSER_OUT3 — J204 U.FL-R-SMT-1_80_
PULSER_OUT4 — J205 U.FL-R-SMT-1_80_
PULSER_OUT5 — J206 U.FL-R-SMT-1_80_
PULSER_OUT6 — J207 U.FL-R-SMT-1_80_
PULSER_OUT7 — J208 U.FL-R-SMT-1_80_
PULSER_OUT8 — J209 U.FL-R-SMT-1_80_
PULSER_OUT9 — J210 U.FL-R-SMT-1_80_
PULSER_OUT10 — J211 U.FL-R-SMT-1_80_
PULSER_OUT11 — J212 U.FL-R-SMT-1_80_
PULSER_OUT12 — J213 U.FL-R-SMT-1_80_
PULSER_OUT13 — J214 U.FL-R-SMT-1_80_
PULSER_OUT14 — J215 U.FL-R-SMT-1_80_
PULSER_OUT15 — J216 U.FL-R-SMT-1_80_
GND

PULSER_100V
C204 100nF 200V
C209 100nF 200V
C212 100nF 200V
C215 100nF 200V
GND

PULSER_50V
C218 100nF 200V
C221 100nF 200V
C224 100nF 200V
C227 100nF 200V
GND

PULSER_100VN
C205 100nF 200V
C210 100nF 200V
C213 100nF 200V
C216 100nF 200V
GND

PULSER_50VN
C219 100nF 200V
C222 100nF 200V
C225 100nF 200V
C228 100nF 200V
GND

PULSER_5V
C206 100nF
C211 100nF
C214 100nF
C217 100nF
GND

PULSER_5VN
C220 100nF
C223 100nF
C226 100nF
C229 100nF
GND

PULSER_1V8
C207 100nF
C208 100nF
GND

Fanout Buffer 1

Feeds:
SOM sync signal
Pulser beamformer sync signal
AFE fanout buffer –>

Table 8-1. Input Selection Table

IN_SEL	ACTIVE CLOCK INPUT
0	IN0_P, IN0_N
1	IN1_P, IN1_N
Open	None (1)

72 MHz Crystal

Y301
ASFLMB-72.000MHZ-LY-T

R303 50R
R301 1K
R302 1K

Bias for common mode voltage (0.9v)

U301
LMK1D1204RGTR

It was recommended by the datasheet to terminate used & unused channels.

R304 100R
R306 100R
R305 100R

Fanout Buffer 2

Feeds:
4 * AFE

U302
LMK1D1204RGTR

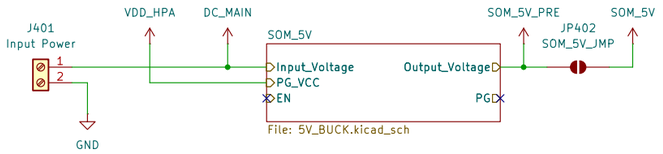
Terminated at AFE side

AFE IO are 3V3

C301 10uF
C302 100nF
C305 100nF
C309 10uF
C314 100nF
C323 10uF
C324 100nF
C327 100nF
C303 1uF
C306 1uF
C310 1uF
C312 1uF
C315 1uF
C317 1uF
C319 1uF
C321 1uF

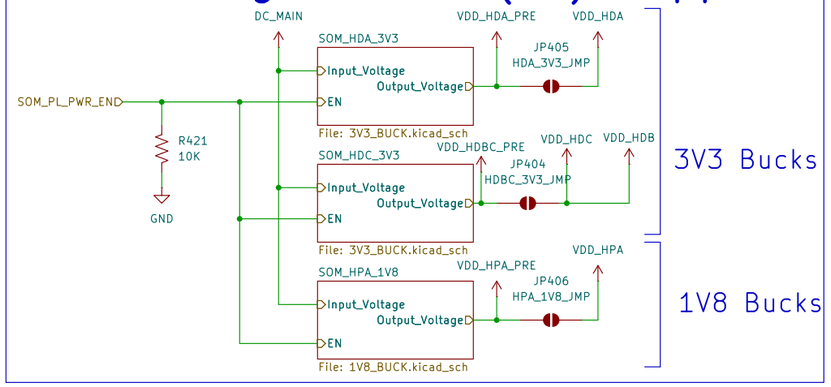
AFE1
AFE2
AFE3
AFE4
File: RX_LANE.kicad_sch

Main SOM 5V Supply

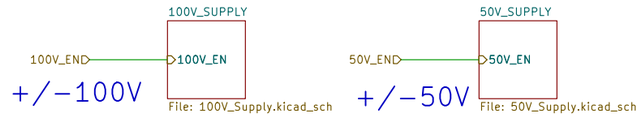


Power Rail Name	Supported Voltage Range	Maximum Current	Description
V_{CC_SOM}	5V (4.75V – 5.25V) 50 mV p-p maximum noise	4A	Main power input to the SOM. Supplies power to on-board power regulators. Total of 4A across connectors. SOM will consume a maximum of 3A.
V_{CC_BATT}	1.20 – 1.50V	[illegible]	Optional battery input for the RTC
V_{CCO_HPA}	1.00V – 1.80V	1.0A	Voltage rail for HPIO bank 66
V_{CCO_HPB}	1.00V – 1.80V	1.0A	Voltage rail for HPIO bank 65
V_{CCO_HPC}	1.00V – 1.80V	1.0A	Voltage rail for HPIO bank 64
V_{CCO_HDA}	1.20V – 3.30V	1.0A	Voltage rail for HDIO bank 45
V_{CCO_HDB}	1.20V – 3.30V	1.0A	Voltage rail for HDIO bank 43
V_{CCO_HDC}	1.20V – 3.30V	1.0A	Voltage rail for HDIO bank 44

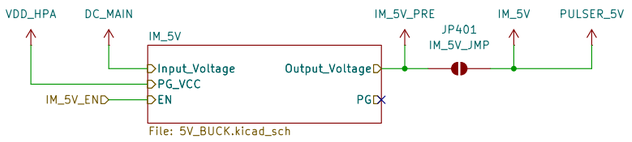
Processing Section (PL) Supplies



HV Supplies (+/- 50/100V)



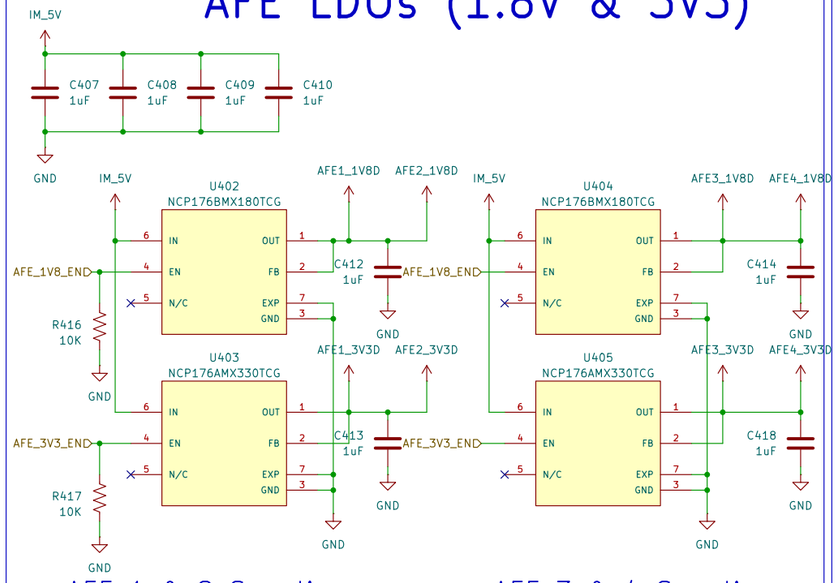
Pulser + Intermediate 5V Supply



I_{VDD18}	Full channel mode	130	mA
I_{VDD33}	Full channel mode	130	mA
I_{VDD18}		22	mA
I_{VDD33}		2	mA

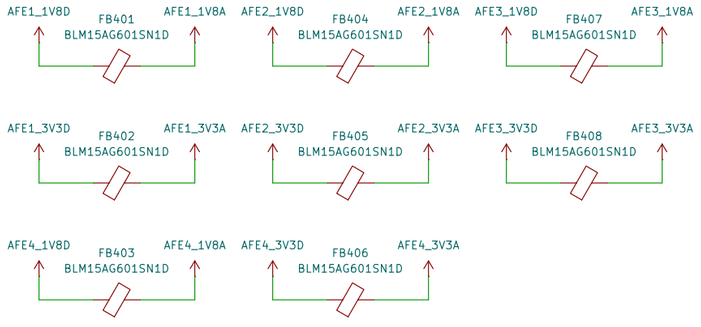
^ x4 = 1.2A

AFE LDOs (1.8V & 3V3)

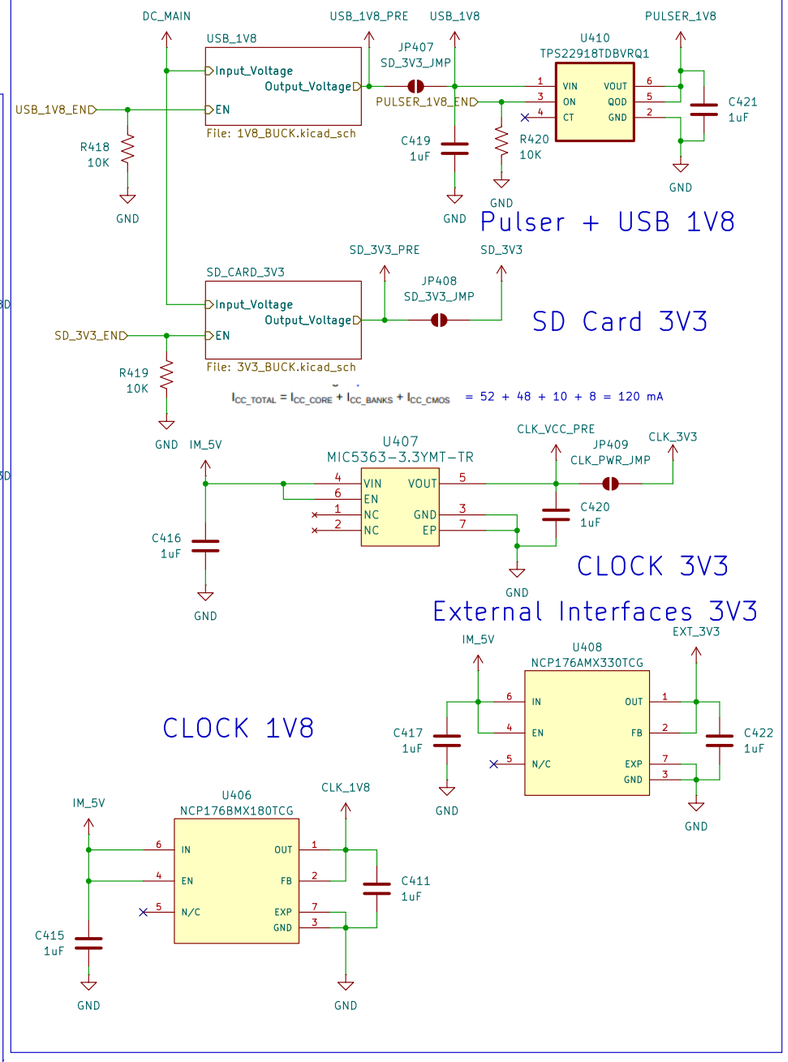


AFE 1 & 2 Supplies

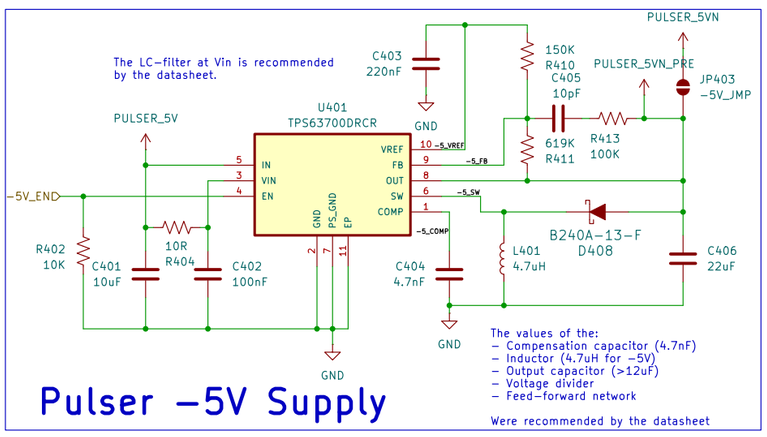
AFE 3 & 4 Supplies

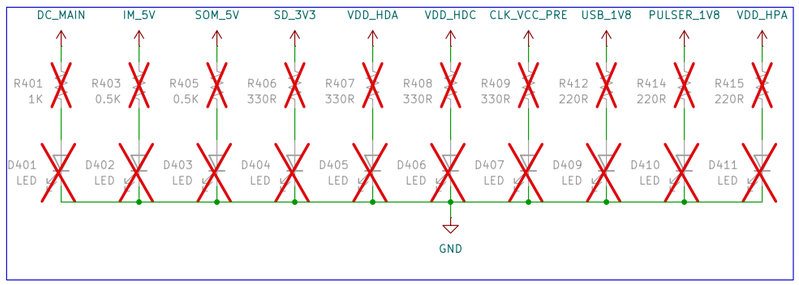


Miscellaneous Supplies



Pulser −5V Supply





VCC_BATT	PS BBRAM and real-time clock (RTC) supply voltage, requires external battery. Connect to GND when battery is not used.

J501A
REF-226081-01

J501B
REF-226081-01

J502A
REF-226081-01

J502B
REF-226081-01

SOM240_1

SOM240_2

HPA
HDA
HDB
HDC
Bank 501
Bank 502

PULSER_SPI_D0_P
PULSER_SPI_D0_N
PULSER_SPI_CLK_P
PULSER_SPI_CLK_N
PULSER_SPI_D1_P
PULSER_SPI_D1_N
PULSER_SPI_SE_P
PULSER_SPI_SE_N
PULSER_SPI_CS0
PULSER_SPI_CS1
PULSER_BF_SYNC_CLK_P
PULSER_BF_SYNC_CLK_N
PULSER_SPI_MODE
PULSER_SPI_CRC_EN
PULSER_CRC_ERR
PULSER_WAKEUP
PULSER_TSHUT
PULSER_RESET
PULSER_1V8_EN
PULSER_TXRX
PULSER_SPI_SDOUT
-5V_EN

AFE1_AUX
AFE1_CSn
AFE1_DSYNC
AFE1_MUXA
AFE2_CSn
AFE2_DSYNC
AFE3_AUX
AFE3_DSYNC
AFE4_MUXA
AFE2_AUX
AFE3_MUXA
AFE3_CSn
AFE4_AUX
AFE4_CSn
AFE4_DSYNC
AFE_SCLK
AFE2_MUXA
AFE_SDIO
SD_DETECT
EXT_INT1
EXT_INT2

SOM_PS_PWR_EN
SOM_PL_PWR_EN
SOM_SYNC_CLK_P
SOM_SYNC_CLK_N
VCC_SOM
R501
10K
PWROFF_ C2M_L

JTAG_TMS_C2M
JTAG_TDO_M2C
JTAG_TDI_C2M
JTAG_TCK_C2M

BOOT_MODE0
BOOT_MODE1
BOOT_MODE2
BOOT_MODE3

LCD_TX_MAIN
LCD_RX_MAIN
LCD_POWER

USB_D0
USB_D1
USB_D2
USB_D3
USB_D4
USB_D5
USB_D6
USB_D7
USB_STP
50V_EN
USB_1V8_EN
EXT_MISO
USB_NXT
AFE_1V8_EN
AFE_3V3_EN
USB_CLK
EXT_AFE_CS1
EXT_AFE_CS2

SD_DAT0
SD_DAT1
SD_DAT2
SD_DAT3
SD_CLK
SD_CMD
USB_RESET
USB_DIR_CTRL
IM_5V_EN
EXT_RESET
EXT_SCLK
EXT_MUX_CS
EXT_MOSI
EXT_MUX_CS

SD_3V3_EN
100V_EN

VDD_HPA
VDD_HDA
VDD_HPB
VDD_HDB
VDD_HPC
VDD_HDC
SOM_5V
GND

AFE_OUTPUTS[0..47]
AFE_OUTPUTS0
AFE_OUTPUTS1
AFE_OUTPUTS2
AFE_OUTPUTS3
AFE_OUTPUTS4
AFE_OUTPUTS5
AFE_OUTPUTS6
AFE_OUTPUTS7
AFE_OUTPUTS8
AFE_OUTPUTS9
AFE_OUTPUTS10
AFE_OUTPUTS11
AFE_OUTPUTS12
AFE_OUTPUTS13
AFE_OUTPUTS14
AFE_OUTPUTS15
AFE_OUTPUTS16
AFE_OUTPUTS17
AFE_OUTPUTS18
AFE_OUTPUTS19
AFE_OUTPUTS20
AFE_OUTPUTS21
AFE_OUTPUTS22
AFE_OUTPUTS23
AFE_OUTPUTS24
AFE_OUTPUTS25
AFE_OUTPUTS26
AFE_OUTPUTS27
AFE_OUTPUTS28
AFE_OUTPUTS29
AFE_OUTPUTS30
AFE_OUTPUTS31
AFE_OUTPUTS32
AFE_OUTPUTS33
AFE_OUTPUTS34
AFE_OUTPUTS35
AFE_OUTPUTS36
AFE_OUTPUTS37
AFE_OUTPUTS38
AFE_OUTPUTS39
AFE_OUTPUTS40
AFE_OUTPUTS41
AFE_OUTPUTS42
AFE_OUTPUTS43
AFE_OUTPUTS44
AFE_OUTPUTS45
AFE_OUTPUTS46
AFE_OUTPUTS47

JP501 BM0
JP502 BM1
JP503 BM2
JP504 BM3

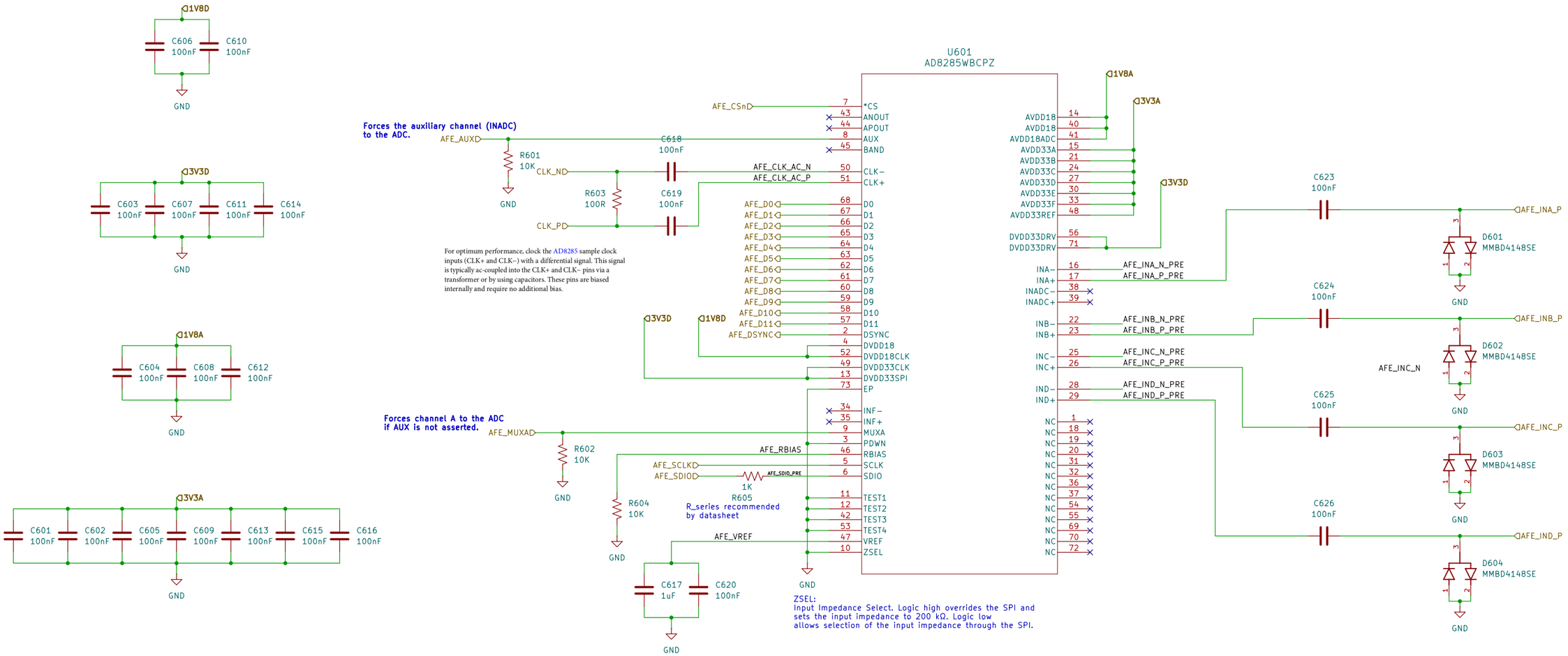
Boot Mode	PS_Mode Pins[3:0]	Physical Pin Location
Quad-SPI (32 bit)	0010	MIO[5:0]
eMMC	0110	MIO[22:13]

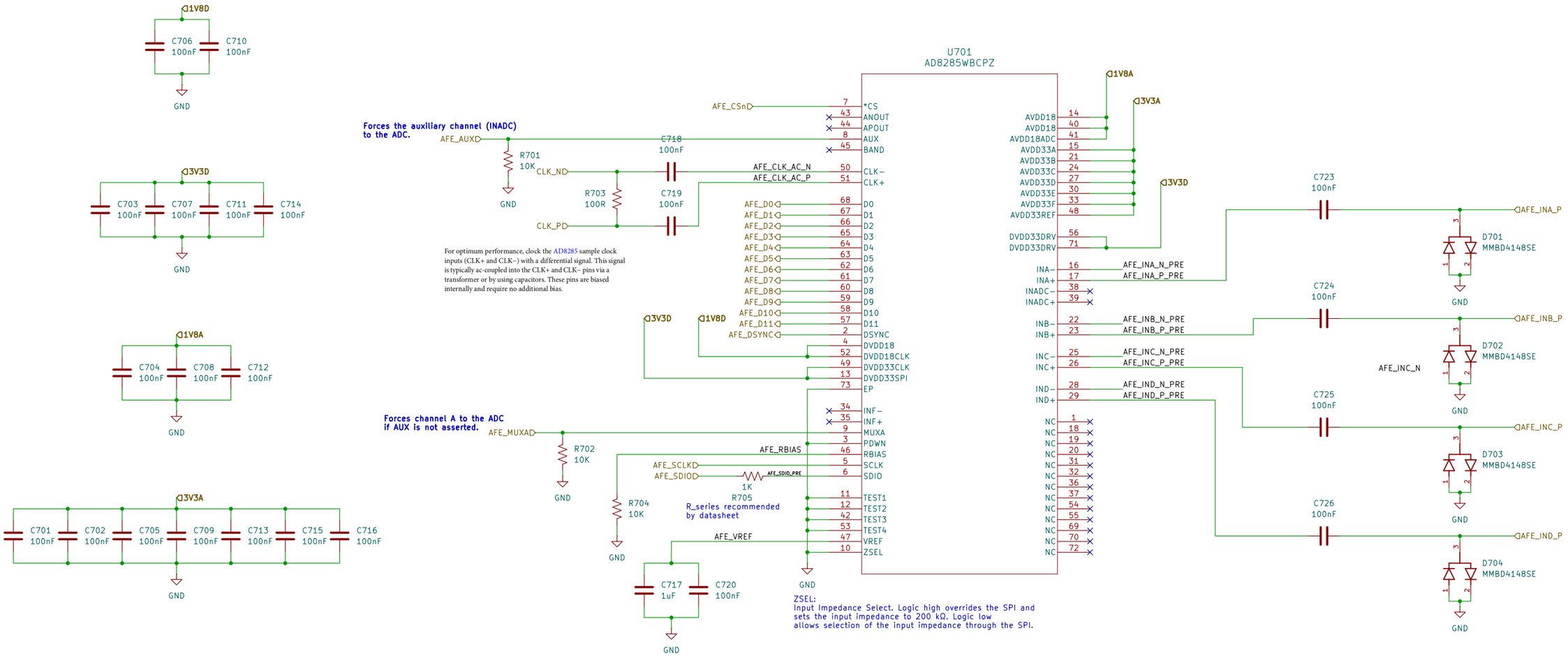
SW501
RESET
C512
100nF

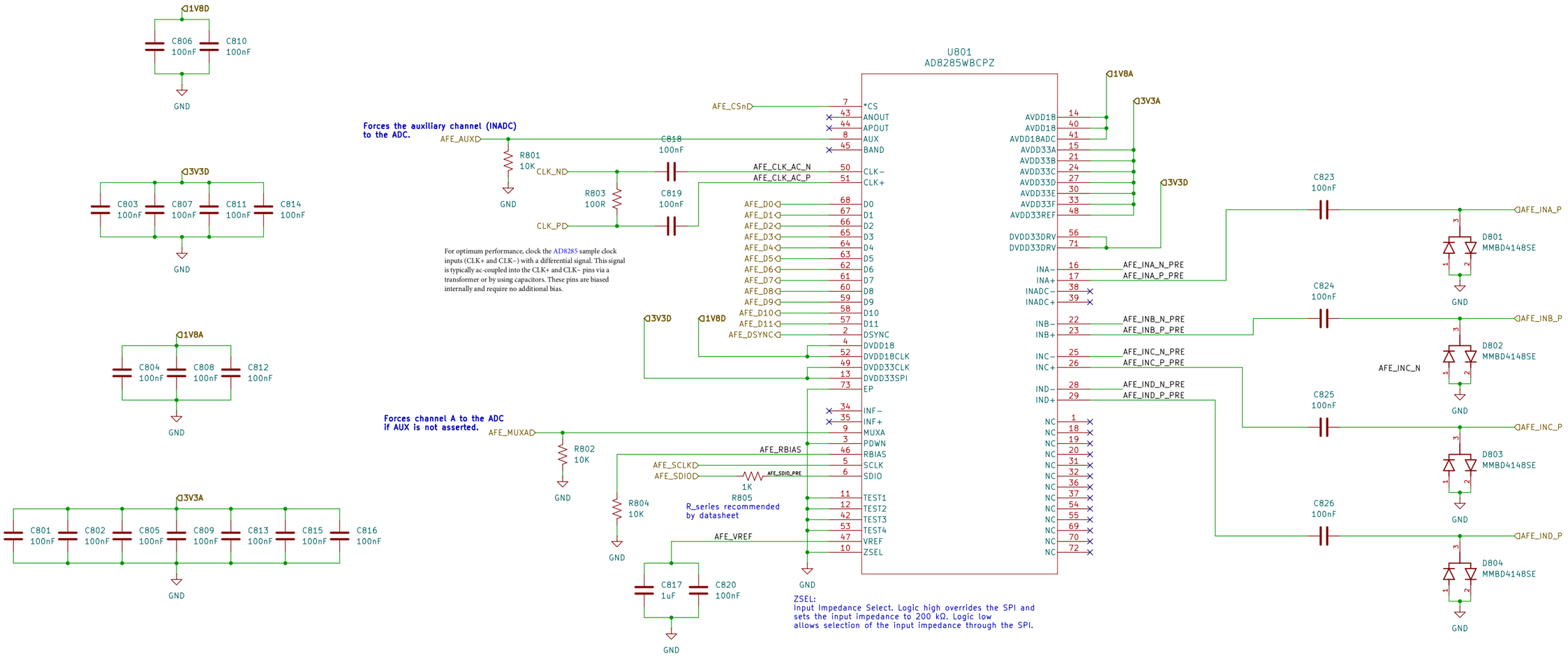
Reset Button

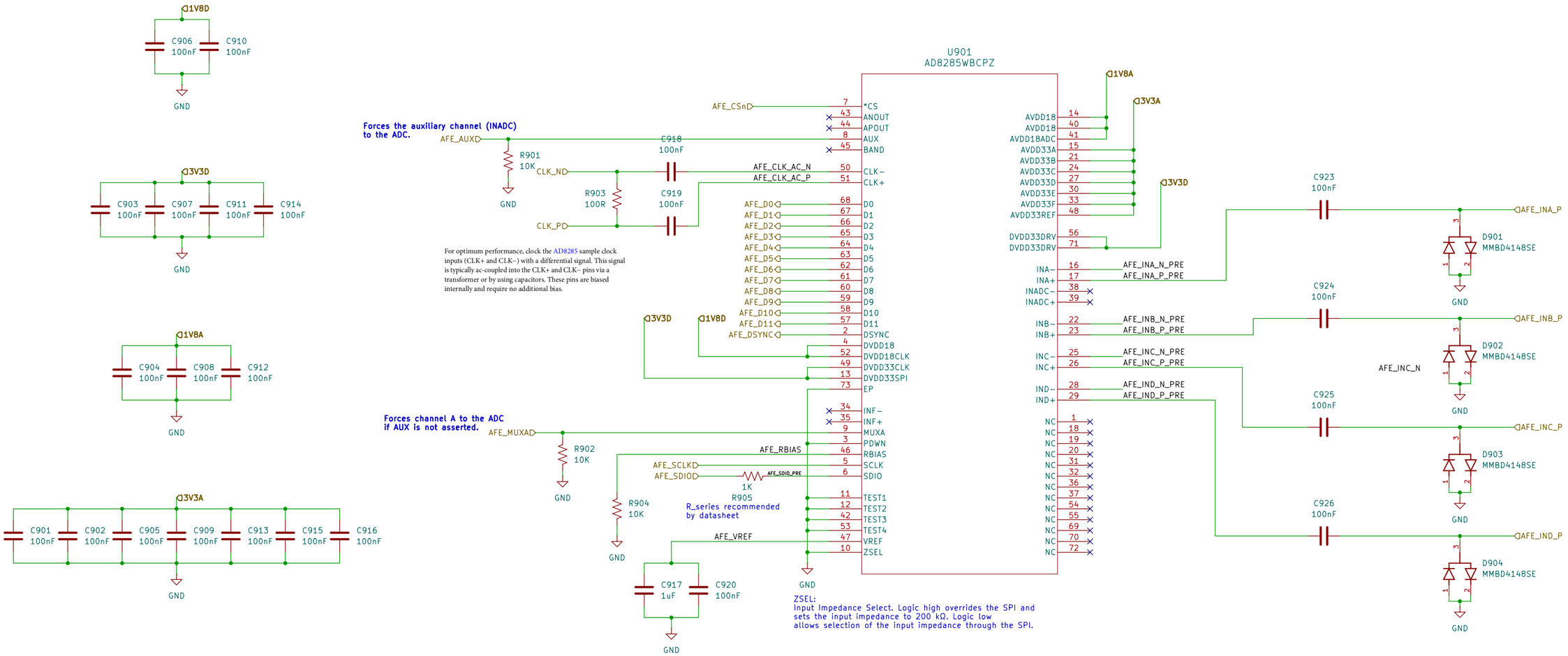
J503
JTAG Header

C501 100nF
C502 1uF
C503 47uF
C504 47uF
C505 100nF
C506 1uF
C507 47uF
C508 47uF
C509 4.7uF
C510 100nF
C511 100nF
C513 4.7uF
C514 100nF
C515 100nF
C516 4.7uF
C517 100nF
C518 100nF
C519 4.7uF
C520 100nF
C521 100nF


1V8D
C606 100nF
C610 100nF
GND
3V3D
C603 100nF
C607 100nF
C611 100nF
C614 100nF
GND
1V8A
C604 100nF
C608 100nF
C612 100nF
GND
3V3A
C601 100nF
C602 100nF
C605 100nF
C609 100nF
C613 100nF
C615 100nF
C616 100nF
GND
U601
AD8285WBCPZ
Forces the auxiliary channel (INADC) to the ADC.
AFE_AUX
R601 10K
GND
C618 100nF
CLK_N
R603 100R
C619 100nF
CLK_P
AFE_CSn
AFE_CLK_AC_N
AFE_CLK_AC_P
AFE_D0
AFE_D1
AFE_D2
AFE_D3
AFE_D4
AFE_D5
AFE_D6
AFE_D7
AFE_D8
AFE_D9
AFE_D10
AFE_D11
AFE_DSYNC
*CS
ANOUT
APOUT
AUX
BAND
CLK−
CLK+
D0
D1
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11
DSYNC
DVDD18
DVDD18CLK
DVDD33CLK
DVDD33SPI
EP
For optimum performance, clock the AD8285 sample clock inputs (CLK+ and CLK−) with a differential signal. This signal is typically ac-coupled into the CLK+ and CLK− pins via a transformer or by using capacitors. These pins are biased internally and require no additional bias.
3V3D
1V8D
Forces channel A to the ADC if AUX is not asserted.
AFE_MUXA
R602 10K
GND
AFE_RBIAS
AFE_SCLK
AFE_SDIO
AFE_SDIO_PRE
1K
R605
R_series recommended by datasheet
R604 10K
GND
INF−
INF+
MUXA
PDWN
RBIAS
SCLK
SDIO
TEST1
TEST2
TEST3
TEST4
VREF
ZSEL
AFE_VREF
C617 1uF
C620 100nF
GND
GND
ZSEL:
Input Impedance Select. Logic high overrides the SPI and sets the input impedance to 200 kΩ. Logic low allows selection of the input impedance through the SPI.
1V8A
3V3A
AVDD18
AVDD18
AVDD18ADC
AVDD33A
AVDD33B
AVDD33C
AVDD33D
AVDD33E
AVDD33F
AVDD33REF
3V3D
DVDD33DRV
DVDD33DRV
INA−
INA+
INADC−
INADC+
INB−
INB+
INC−
INC+
IND−
IND+
AFE_INA_N_PRE
AFE_INA_P_PRE
AFE_INB_N_PRE
AFE_INB_P_PRE
AFE_INC_N_PRE
AFE_INC_P_PRE
AFE_IND_N_PRE
AFE_IND_P_PRE
NC
C623 100nF
C624 100nF
C625 100nF
C626 100nF
D601 MMBD4148SE
D602 MMBD4148SE
D603 MMBD4148SE
D604 MMBD4148SE
AFE_INA_P
AFE_INB_P
AFE_INC_P
AFE_IND_P
AFE_INC_N
GND
AFE_INA_N_PRE
AFE_INB_N_PRE
AFE_INC_N_PRE
AFE_IND_N_PRE
R606 8R
R607 8R
R608 8R
R609 8R
Change value according to pulser T/R switch out impedance
C621 100nF
C622 100nF
C627 100nF
C628 100nF
GND


1V8D
C706 100nF
C710 100nF
GND
3V3D
C703 100nF
C707 100nF
C711 100nF
C714 100nF
GND
1V8A
C704 100nF
C708 100nF
C712 100nF
GND
3V3A
C701 100nF
C702 100nF
C705 100nF
C709 100nF
C713 100nF
C715 100nF
C716 100nF
GND
U701
AD8285WBCPZ
Forces the auxiliary channel (INADC) to the ADC.
AFE_AUX
R701 10K
GND
C718 100nF
CLK_N
R703 100R
C719 100nF
CLK_P
AFE_CLK_AC_N
AFE_CLK_AC_P
For optimum performance, clock the AD8285 sample clock inputs (CLK+ and CLK−) with a differential signal. This signal is typically ac-coupled into the CLK+ and CLK− pins via a transformer or by using capacitors. These pins are biased internally and require no additional bias.
AFE_CSn
AFE_D0
AFE_D1
AFE_D2
AFE_D3
AFE_D4
AFE_D5
AFE_D6
AFE_D7
AFE_D8
AFE_D9
AFE_D10
AFE_D11
AFE_DSYNC
3V3D
1V8D
Forces channel A to the ADC if AUX is not asserted.
AFE_MUXA
R702 10K
GND
AFE_RBIAS
AFE_SCLK
AFE_SDIO
AFE_SDIO_PRE
1K
R705
R_series recommended by datasheet
R704 10K
GND
AFE_VREF
C717 1uF
C720 100nF
GND
*CS
ANOUT
APOUT
AUX
BAND
CLK−
CLK+
D0
D1
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11
DSYNC
DVDD18
DVDD18CLK
DVDD33CLK
DVDD33SPI
EP
INF−
INF+
MUXA
PDWN
RBIAS
SCLK
SDIO
TEST1
TEST2
TEST3
TEST4
VREF
ZSEL
GND
ZSEL:
Input Impedance Select. Logic high overrides the SPI and sets the input impedance to 200 kΩ. Logic low allows selection of the input impedance through the SPI.
1V8A
3V3A
3V3D
AVDD18
AVDD18
AVDD18ADC
AVDD33A
AVDD33B
AVDD33C
AVDD33D
AVDD33E
AVDD33F
AVDD33REF
DVDD33DRV
DVDD33DRV
INA−
INA+
INADC−
INADC+
INB−
INB+
INC−
INC+
IND−
IND+
AFE_INA_N_PRE
AFE_INA_P_PRE
AFE_INB_N_PRE
AFE_INB_P_PRE
AFE_INC_N_PRE
AFE_INC_P_PRE
AFE_IND_N_PRE
AFE_IND_P_PRE
NC
C723 100nF
C724 100nF
C725 100nF
C726 100nF
D701 MMBD4148SE
D702 MMBD4148SE
D703 MMBD4148SE
D704 MMBD4148SE
AFE_INA_P
AFE_INB_P
AFE_INC_N
AFE_INC_P
AFE_IND_P
GND
AFE_INA_N_PRE
AFE_INB_N_PRE
AFE_INC_N_PRE
AFE_IND_N_PRE
R706 8R
R707 8R
R708 8R
R709 8R
Change value according to pulser T/R switch out impedance
C721 100nF
C722 100nF
C727 100nF
C728 100nF
GND

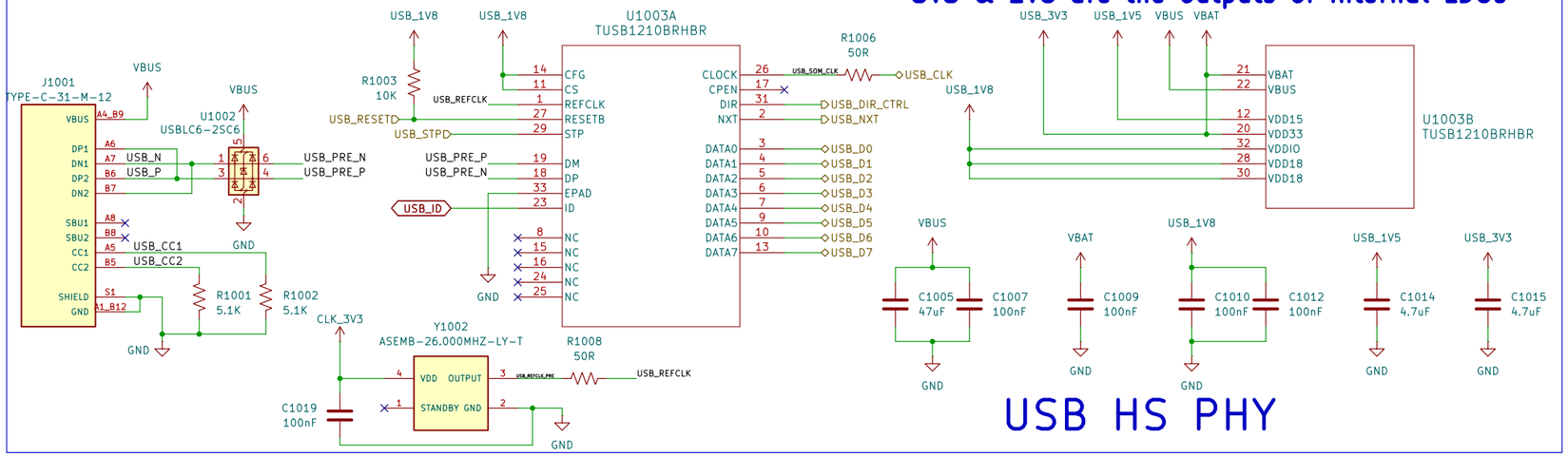

1V8D
C806 100nF
C810 100nF
GND
3V3D
C803 100nF
C807 100nF
C811 100nF
C814 100nF
GND
1V8A
C804 100nF
C808 100nF
C812 100nF
GND
3V3A
C801 100nF
C802 100nF
C805 100nF
C809 100nF
C813 100nF
C815 100nF
C816 100nF
GND
U801
AD8285WBCPZ
Forces the auxiliary channel (INADC) to the ADC.
AFE_AUX
C818 100nF
R801 10K
GND
CLK_N
R803 100R
C819 100nF
CLK_P
AFE_CSn
AFE_CLK_AC_N
AFE_CLK_AC_P
AFE_D0
AFE_D1
AFE_D2
AFE_D3
AFE_D4
AFE_D5
AFE_D6
AFE_D7
AFE_D8
AFE_D9
AFE_D10
AFE_D11
AFE_DSYNC
*CS
ANOUT
APOUT
AUX
BAND
CLK−
CLK+
D0
D1
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11
DSYNC
DVDD18
DVDD18CLK
DVDD33CLK
DVDD33SPI
EP
For optimum performance, clock the AD8285 sample clock inputs (CLK+ and CLK−) with a differential signal. This signal is typically ac-coupled into the CLK+ and CLK− pins via a transformer or by using capacitors. These pins are biased internally and require no additional bias.
3V3D
1V8D
Forces channel A to the ADC if AUX is not asserted.
AFE_MUXA
R802 10K
GND
AFE_RBIAS
AFE_SCLK
AFE_SDIO
AFE_SDIO_PRE
1K
R805
R804 10K
R_series recommended by datasheet
GND
INF−
INF+
MUXA
PDWN
RBIAS
SCLK
SDIO
TEST1
TEST2
TEST3
TEST4
VREF
ZSEL
AFE_VREF
C817 1uF
C820 100nF
GND
ZSEL:
Input Impedance Select. Logic high overrides the SPI and sets the input impedance to 200 kΩ. Logic low allows selection of the input impedance through the SPI.
1V8A
3V3A
3V3D
AVDD18
AVDD18
AVDD18ADC
AVDD33A
AVDD33B
AVDD33C
AVDD33D
AVDD33E
AVDD33F
AVDD33REF
DVDD33DRV
DVDD33DRV
INA−
INA+
INADC−
INADC+
INB−
INB+
INC−
INC+
IND−
IND+
NC
AFE_INA_N_PRE
AFE_INA_P_PRE
AFE_INB_N_PRE
AFE_INB_P_PRE
AFE_INC_N_PRE
AFE_INC_P_PRE
AFE_IND_N_PRE
AFE_IND_P_PRE
C823 100nF
C824 100nF
C825 100nF
C826 100nF
AFE_INA_P
AFE_INB_P
AFE_INC_P
AFE_IND_P
AFE_INC_N
D801 MMBD4148SE
D802 MMBD4148SE
D803 MMBD4148SE
D804 MMBD4148SE
GND
AFE_INA_N_PRE
AFE_INB_N_PRE
AFE_INC_N_PRE
AFE_IND_N_PRE
R806 8R
R807 8R
R808 8R
R809 8R
Change value according to pulser T/R switch out impedance
C821 100nF
C822 100nF
C827 100nF
C828 100nF
GND


1V8D
C906 100nF
C910 100nF
GND
3V3D
C903 100nF
C907 100nF
C911 100nF
C914 100nF
GND
1V8A
C904 100nF
C908 100nF
C912 100nF
GND
3V3A
C901 100nF
C902 100nF
C905 100nF
C909 100nF
C913 100nF
C915 100nF
C916 100nF
GND
U901
AD8285WBCPZ
Forces the auxiliary channel (INADC) to the ADC.
AFE_AUX
R901 10K
GND
C918 100nF
CLK_N
R903 100R
C919 100nF
CLK_P
AFE_CSn
AFE_CLK_AC_N
AFE_CLK_AC_P
AFE_D0
AFE_D1
AFE_D2
AFE_D3
AFE_D4
AFE_D5
AFE_D6
AFE_D7
AFE_D8
AFE_D9
AFE_D10
AFE_D11
AFE_DSYNC
*CS
ANOUT
APOUT
AUX
BAND
CLK−
CLK+
D0
D1
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11
DSYNC
DVDD18
DVDD18CLK
DVDD33CLK
DVDD33SPI
EP
For optimum performance, clock the AD8285 sample clock inputs (CLK+ and CLK−) with a differential signal. This signal is typically ac-coupled into the CLK+ and CLK− pins via a transformer or by using capacitors. These pins are biased internally and require no additional bias.
3V3D
1V8D
Forces channel A to the ADC if AUX is not asserted.
AFE_MUXA
R902 10K
GND
AFE_RBIAS
AFE_SCLK
AFE_SDIO
AFE_SDIO_PRE
1K
R905
R904 10K
R_series recommended by datasheet
GND
INF−
INF+
MUXA
PDWN
RBIAS
SCLK
SDIO
TEST1
TEST2
TEST3
TEST4
VREF
ZSEL
AFE_VREF
C917 1uF
C920 100nF
GND
ZSEL:
Input Impedance Select. Logic high overrides the SPI and sets the input impedance to 200 kΩ. Logic low allows selection of the input impedance through the SPI.
1V8A
3V3A
3V3D
AVDD18
AVDD18
AVDD18ADC
AVDD33A
AVDD33B
AVDD33C
AVDD33D
AVDD33E
AVDD33F
AVDD33REF
DVDD33DRV
DVDD33DRV
INA−
INA+
INADC−
INADC+
INB−
INB+
INC−
INC+
IND−
IND+
NC
AFE_INA_N_PRE
AFE_INA_P_PRE
AFE_INB_N_PRE
AFE_INB_P_PRE
AFE_INC_N_PRE
AFE_INC_P_PRE
AFE_IND_N_PRE
AFE_IND_P_PRE
C923 100nF
C924 100nF
C925 100nF
C926 100nF
AFE_INA_P
AFE_INB_P
AFE_INC_P
AFE_IND_P
AFE_INC_N
D901 MMBD4148SE
D902 MMBD4148SE
D903 MMBD4148SE
D904 MMBD4148SE
GND
AFE_INA_N_PRE
AFE_INB_N_PRE
AFE_INC_N_PRE
AFE_IND_N_PRE
R906 8R
R907 8R
R908 8R
R909 8R
Change value according to pulser T/R switch out impedance
C921 100nF
C922 100nF
C927 100nF
C928 100nF
GND

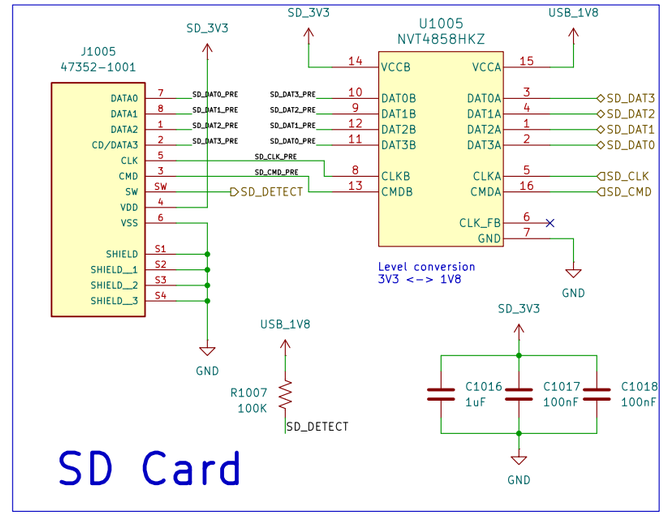
Check AN1310 for layout guidelines

CFG	14	D	I	V_{DDIO}	REFCLK clock frequency configuration pin. Two frequencies are supported: 19.2 MHz when 0, or 26 MHz when 1.
CS	11	D	I	V_{DDIO}	Active-high chip select pin. When low the IC is in power down and ULPI bus is tri-stated. When high normal operation. Tie to V_{DDIO} if unused.

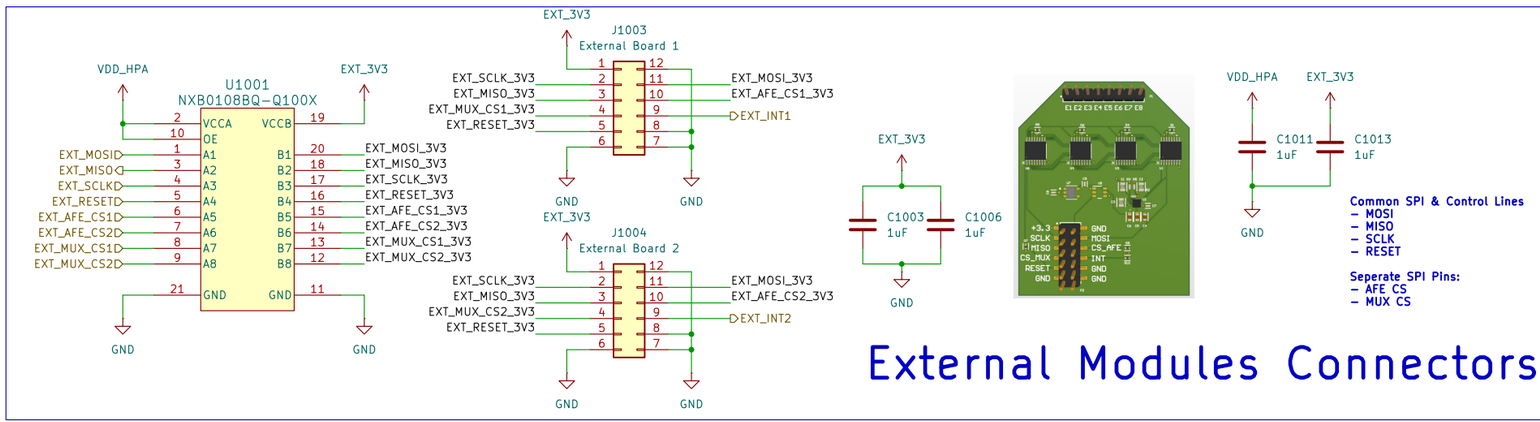
3V3 & 1V5 are the outputs of internal LDOs



USB HS PHY



SD Card

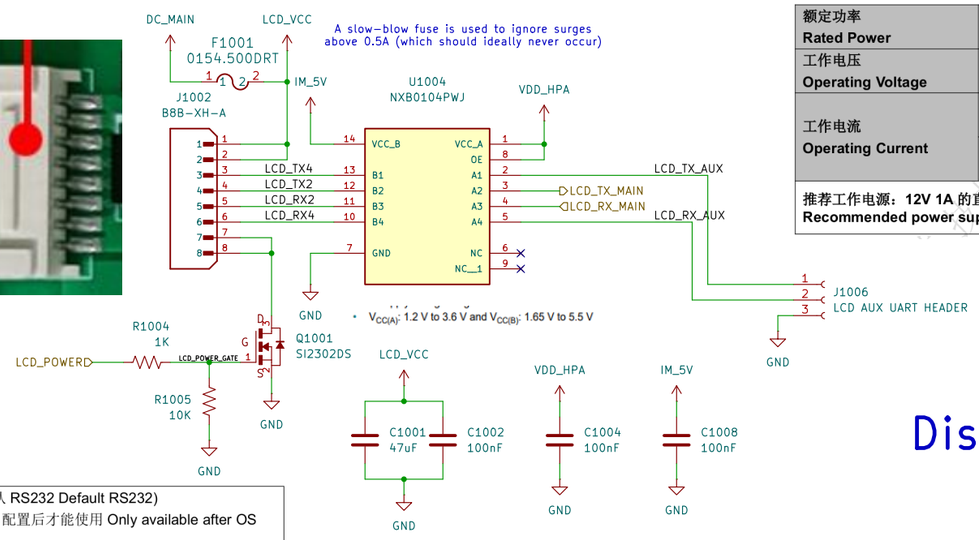


Common SPI & Control Lines
- MOSI
- MISO
- SCLK
- RESET

Seperate SPI Pins:
- AFE CS
- MUX CS

External Modules Connectors



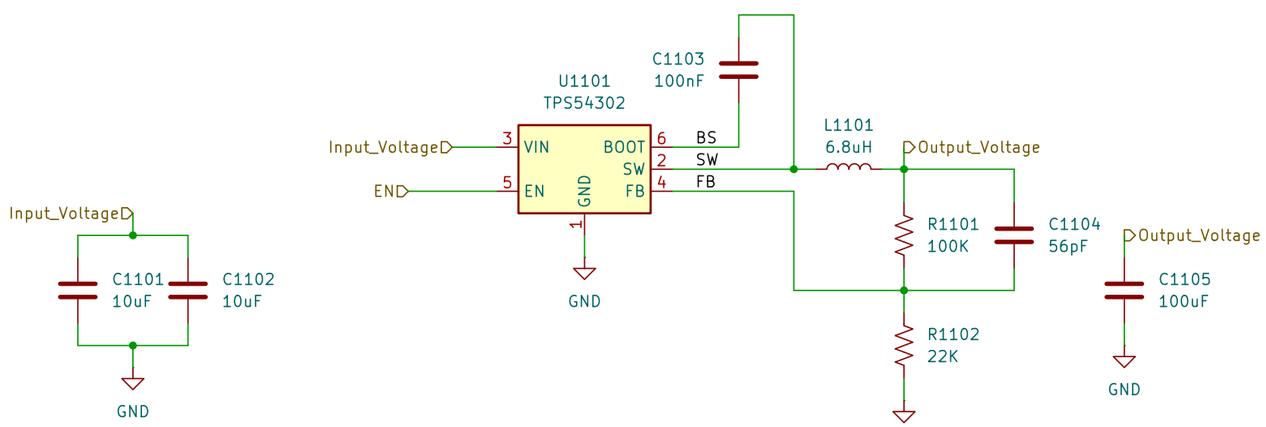


额定功率 **Rated Power**	<5W	
工作电压 **Operating Voltage**	9~36V，典型值 12V 9~36V, typical value of 12V	
工作电流 **Operating Current**	320mA	VCC=12V，背光亮度最大 VCC=12V, max backlight
	120mA	VCC=12V，背光关闭 VCC=12V, backlight off

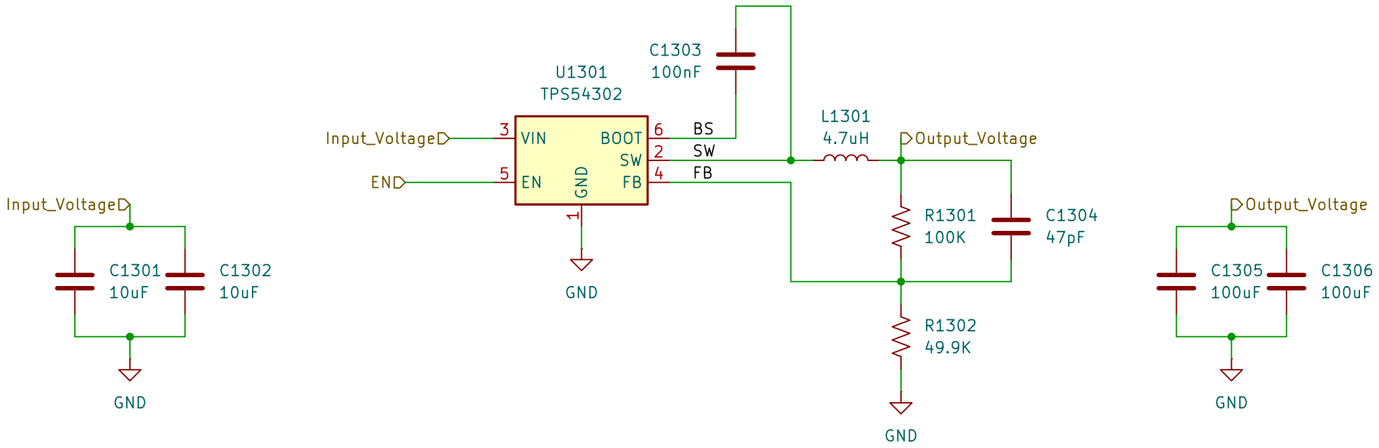
推荐工作电源：12V 1A 的直流稳压电源
Recommended power supply: 12V 1A DC

Display

UART2: ON=TTL/CMOS; OFF=RS232 (默认 RS232 Default RS232)
UART4: ON=TTL/CMOS; OFF=RS232 (OS 配置后才能使用 Only available after OS configuration)



WEBENCH DESIGNER LINK (3.3V)



WEBENCH DESIGNER LINK (1.8V)

Duty Cycle

$$D_{max} = \frac{V_{out} + V_d}{V_{inmin} + V_{out} + V_d}$$ = (100+0.7) / (11+100+0.7) = 90%
= (50+0.7) / (11+50+0.7) = 82%

F_switching = 150 KHz

Inductor

$$L1 = L2 = L3 = \frac{V_{inmin} \times V_{inmax} \times D_{max}}{2 \times I_{out} \times V_{out} \times 0.4 \times f_{SW}}$$ = (11*25*0.82) / (2*0.3*50*0.4*150000) = 150uH

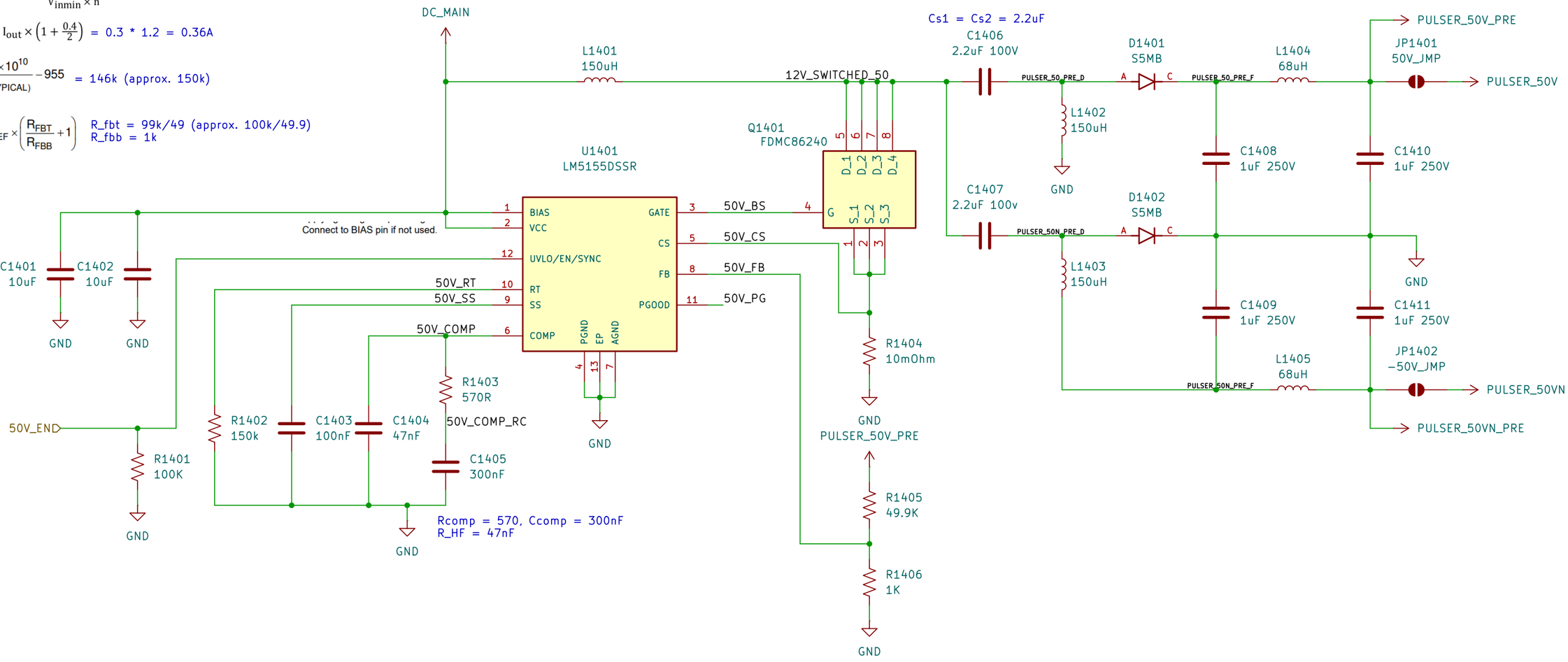
n = 0.88

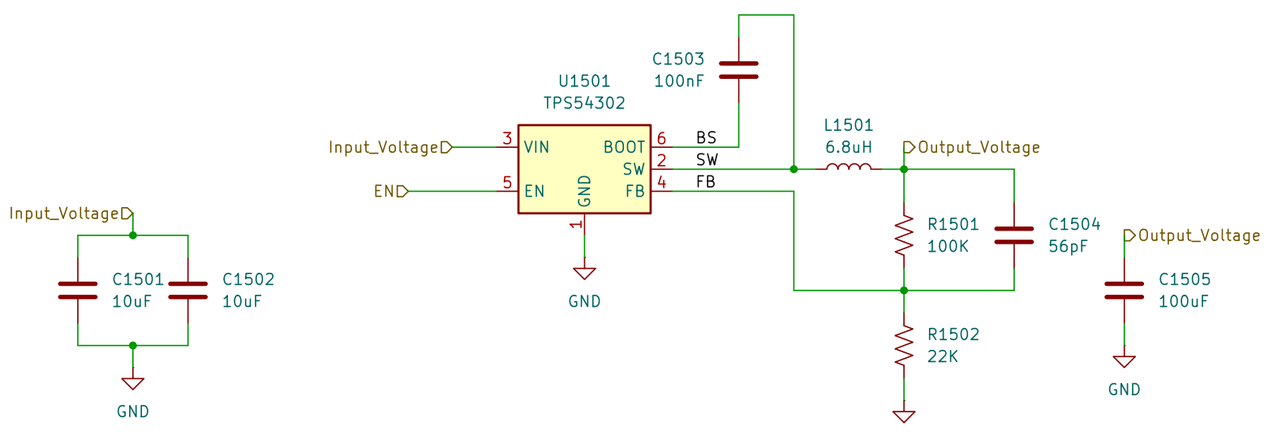
$$I(L1_{peak}) = \frac{I_{out} \times (V_{out} + V_d) \times \left(1 + \frac{0.4}{2}\right)}{V_{inmin} \times n}$$ = (0.3*100.7*1.2) / (11*0.88) = 3.75A

$$I(L2_{peak}) = I_{out} \times \left(1 + \frac{0.4}{2}\right)$$ = 0.3 * 1.2 = 0.36A

$$R_T = \frac{2.21 \times 10^{10}}{f_{RT(TYPICAL)}} - 955$$ = 146k (approx. 150k)

$$V_{LOAD} = V_{REF} \times \left(\frac{R_{FBT}}{R_{FBB}} + 1\right)$$ R_fbt = 99k/49 (approx. 100k/49.9)
R_fbb = 1k





WEBENCH DESIGNER LINK (3.3V)

Duty Cycle

$$D_{max} = \frac{V_{out} + V_d}{V_{inmin} + V_{out} + V_d}$$ = (100+0.7) / (11+100+0.7) = 90%

= (50+0.7) / (11+50+0.7) = 82%

F_switching = 150 KHz

Inductor

$$L1 = L2 = L3 = \frac{V_{inmin} \times V_{inmax} \times D_{max}}{2 \times I_{out} \times V_{out} \times 0.4 \times f_{SW}}$$ = (11*25*0.9) / (2*0.3*100*0.4*150000) = 68uH

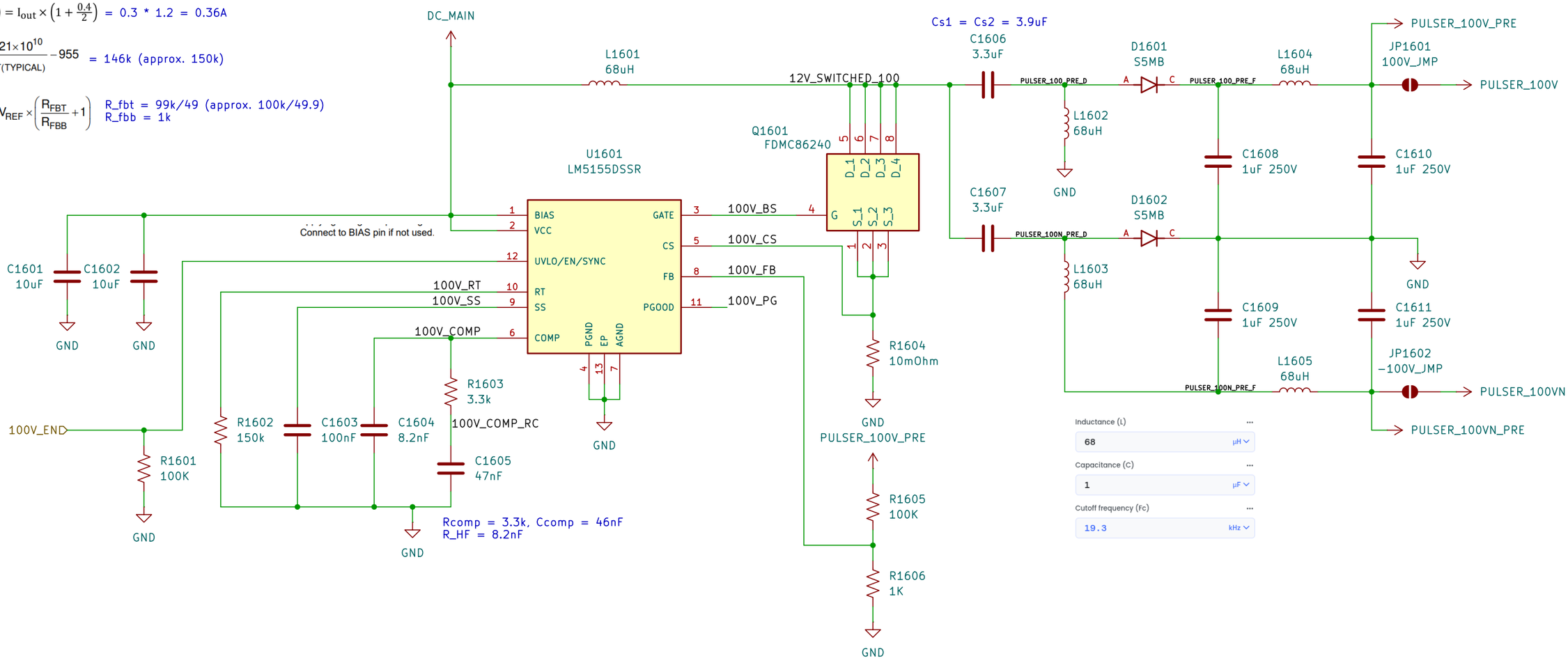
n = 0.88

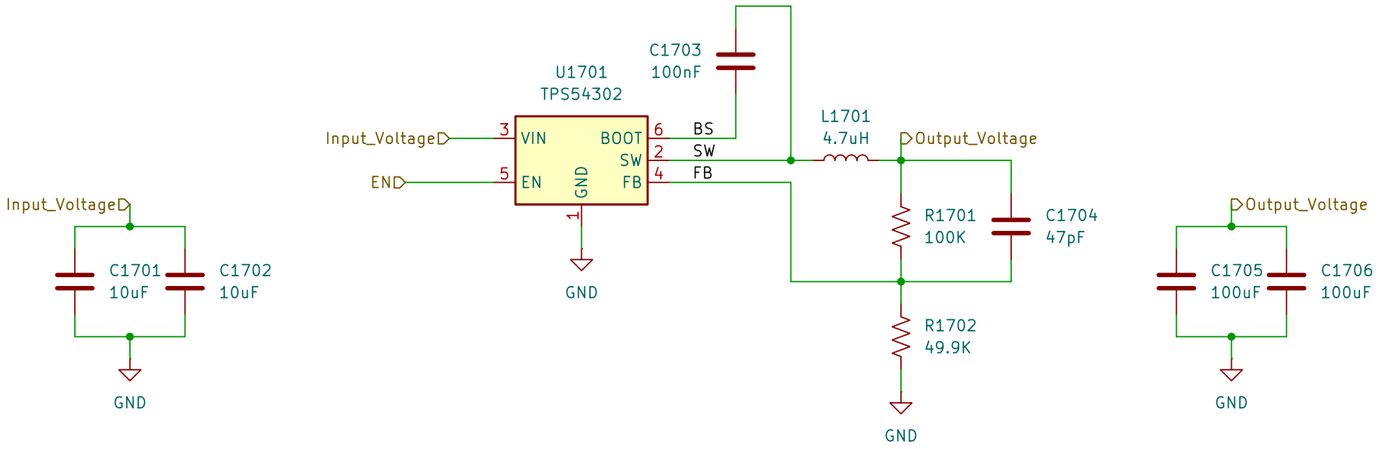
$$I(L1_{peak}) = \frac{I_{out} \times (V_{out} + V_d) \times \left(1 + \frac{0.4}{2}\right)}{V_{inmin} \times n}$$ = (0.3*100.7*1.2) / (11*0.88) = 3.75A

$$I(L2_{peak}) = I_{out} \times \left(1 + \frac{0.4}{2}\right)$$ = 0.3 * 1.2 = 0.36A

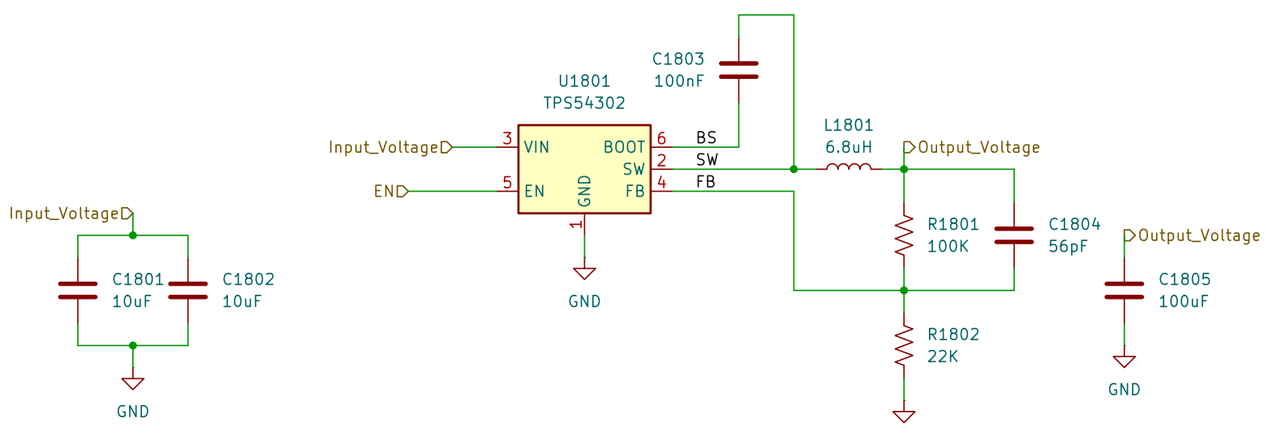
$$R_T = \frac{2.21 \times 10^{10}}{f_{RT(TYPICAL)}} - 955$$ = 146k (approx. 150k)

$$V_{LOAD} = V_{REF} \times \left(\frac{R_{FBT}}{R_{FBB}} + 1\right)$$ R_fbt = 99k/49 (approx. 100k/49.9)
R_fbb = 1k

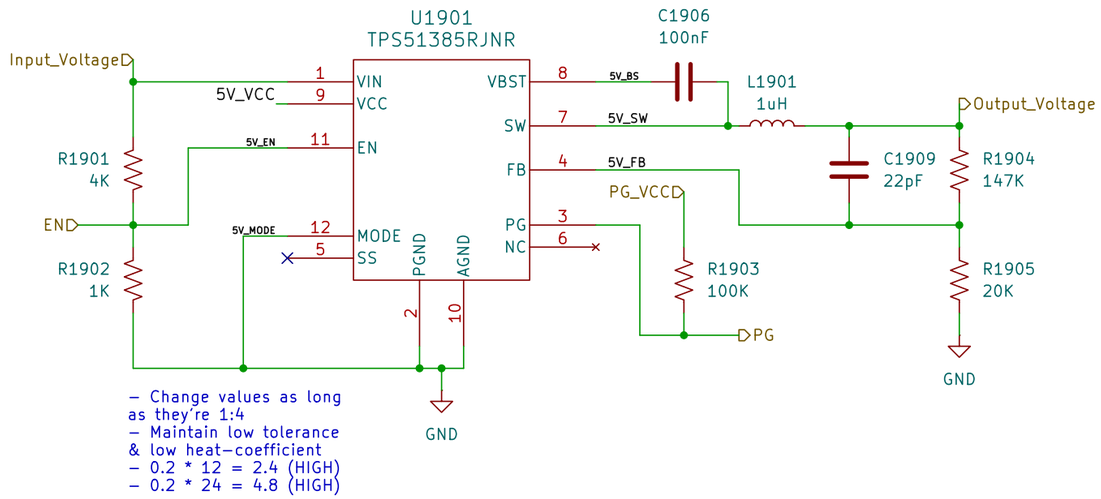


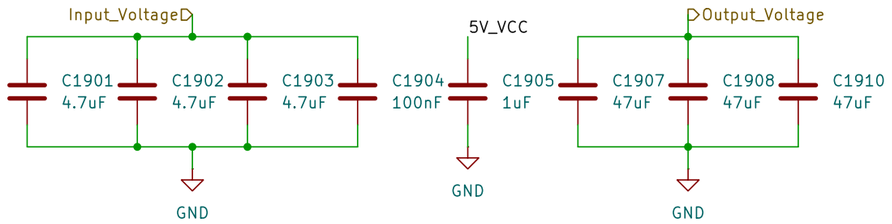


WEBENCH DESIGNER LINK (1.8V)

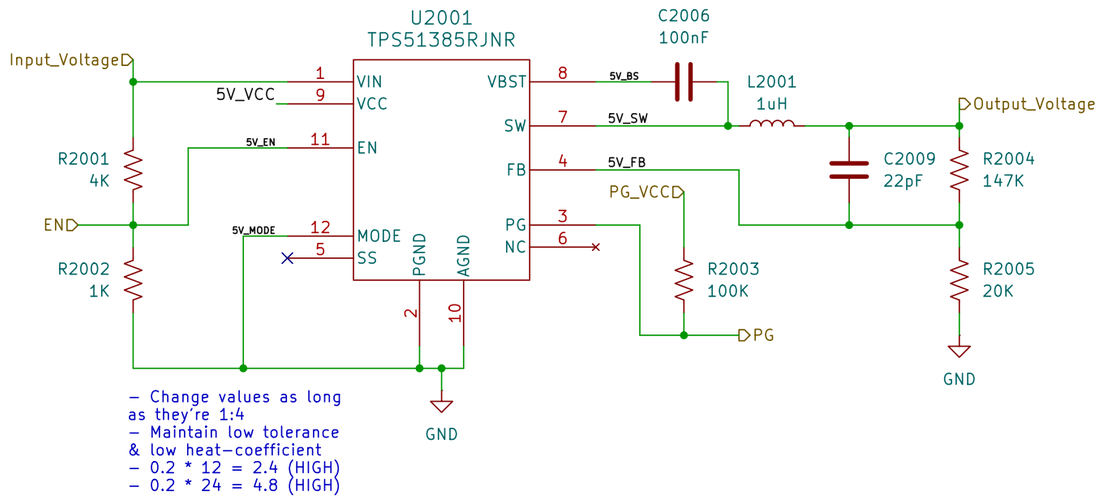


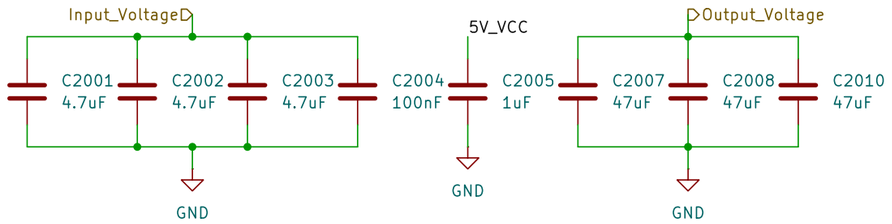
WEBENCH DESIGNER LINK (3.3V)





WeBench Designer Link





WeBench Designer Link